Statistics

Chapter 8 — Instructor Solutions Manual

October 12, 2026 • Total Exercises: 49

PROBLEM STATEMENT

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Step 1

(a) Selected random numbers: 08, 18, 11, 54, 02, 41, 54

The given ID numbers range from 00 to 23, thus we need to eliminate all random numbers that exceed 23:

$$08, 18, 11, 02$$

The corresponding Addresses are then in the sample: 303 Louisiana Av, 5155 S Main, 3501 Monroe St, 2652 W Central Av.

Step 2

(b) Let us select the 11th row of the random numbers table: Note: Choosing a different row will result in a different sample

82173 26921 28472 98958 07960 66124

We select the first 2-digit number. If the number is between 00 and 23, then we select the corresponding location. If the number is greater than 23, then we ignore the number and select the next 2-digit number.

Repeat until 4 unique locations were selected.

82	$\Rightarrow$	Ignore
17	$\Rightarrow$	Select 4624 Woodville Rd
32	$\Rightarrow$	Ignore
69	$\Rightarrow$	Ignore
21	$\Rightarrow$	Select 4252 Monroe
28	$\Rightarrow$	Ignore
47	$\Rightarrow$	Ignore
29	$\Rightarrow$	Ignore
89	$\Rightarrow$	Ignore
58	$\Rightarrow$	Ignore
07	$\Rightarrow$	Select 2525 W Laskey Rd
96	$\Rightarrow$	Ignore
06	$\Rightarrow$	Select 3382 Lagrange Street

Thus the sample will then contain the four locations 4624 Woodville Rd, 4252 Monroe, 2525 W Laskey Rd, 3382 Lagrange Street.

Step 3

(c) We need to select every seventh location with starting point 03.

Let us first determine the Id numbers that will be in the sample, which are the starting point and the starting point increased by 7 or a multiple of 7.

The given ID numbers range from 00 to 23, thus we need to eliminate all random numbers that exceed 23.

$$\begin{aligned} & 03 \\ 03+07= & 10 \\ 10+07= & 17 \\ 17+07= & 24 \end{aligned}$$

The selected random numbers are then 03, 10, and 17. The corresponding locations are: 630 Dixie Hwy, 835 S McCord Rd, 4624 Woodville Rd.

Step 4

(d) We need to select 2 corporate-owned locations and 1 manager-owned location.

Let us select the 11th row of the random numbers table: Note: Choosing a different row will result in a different sample

82173 26921 28472 98958 07960 66124

We select the first 2-digit number. If the number is between 00 and 23, then we select the corresponding location. If the number is greater than 23, then we ignore the number and select the next 2-digit number.

Repeat until 2 corporate-owned locations and 1 manager-owned corporation were selected (while any additional locations will also be ignored).

82	$\Rightarrow$	Ignore
17	$\Rightarrow$	Select 4624 Woodville Rd (M)
32	$\Rightarrow$	Ignore
69	$\Rightarrow$	Ignore
21	$\Rightarrow$	Select 4252 Monroe (C)
28	$\Rightarrow$	Ignore
47	$\Rightarrow$	Ignore
29	$\Rightarrow$	Ignore
89	$\Rightarrow$	Ignore
58	$\Rightarrow$	Ignore
07	$\Rightarrow$	Select 2525 W Laskey Rd (C)

Thus the sample will then contain the three locations 4624 Woodville Rd, 4252 Monroe, 2525 W Laskey Rd.

Result

(a) 303 Louisiana Av, 5155 S Main, 3501 Monroe St, 2652 W Central Av

(b) Answers could vary

(c) 630 Dixie Hwy, 835 S McCord Rd, 4624 Woodville Rd

(d) Answers could vary

PROBLEM STATEMENT

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Step 1

(a) Selected random numbers: 09, 16, 00, 49, 54, 12, 04

The given ID numbers range from 00 to 28, thus we need to eliminate all random numbers that exceed 28:

$$09, 16, 00, 12, 04$$

The corresponding hospitals are then in the sample: Children's Hospital Medical Center, St. Francis-St. George Hospital, Bethesda North, Good Samaritan Hospital, Mercy Hospital-Hamilton.

Step 2

(b) Let us select the 11th row of the random numbers table: Note: Choosing a different row will result in a different sample

82173 26921 28472 98958 07960 66124

We select the first 2-digit number. If the number is between 00 and 28, then we select the corresponding location. If the number is greater than 28, then we ignore the number and select the next 2-digit number.

Repeat until 5 unique hospitals were selected.

82	$\Rightarrow$	Ignore
17	$\Rightarrow$	Select St. Elizabeth Medical Center, North Unit
32	$\Rightarrow$	Ignore
69	$\Rightarrow$	Ignore
21	$\Rightarrow$	Select Care Unit Hospital
28	$\Rightarrow$	Select VA Medical Center Cincinnati
47	$\Rightarrow$	Ignore
29	$\Rightarrow$	Ignore
89	$\Rightarrow$	Ignore
58	$\Rightarrow$	Ignore
07	$\Rightarrow$	Select Mercy Hospital-Anderson
96	$\Rightarrow$	Ignore
06	$\Rightarrow$	Select Clermont Mercy Hospital

Thus the sample will then contain the five hospitals: St. Elizabeth Medical Center North Unit, Care Unit Hospital, VA Medical Center Cincinnati, Mercy Hospital-Anderson, Clermont Mercy Hospital.

Step 3

(c) We need to select every fifth location with starting point 02.

Let us first determine the Id numbers that will be in the sample, which are the starting point and the starting point increased by 5 or a multiple of 5.

The given ID numbers range from 00 to 28, thus we need to eliminate all random numbers that exceed 28.

$$
\begin{aligned}
&02\\
02+05 =&07\\
07+05 =&12\\
12+05 =&17\\
17+05 =&22\\
22+05 =&27\\
27+05 =&32
\end{aligned}
$$

The selected random numbers are then 02, 07, 12, 17, 22, 27. The corresponding hospitals are: Jewish Hospital-Kenwood, Mercy Hospital-Anderson, Good Samaritan Hospital, St. Elizabeth Medical Center North Unit, Emerson Behavioral Science, Shriners Burns Institute.

Step 4

(d) We need to select 4 medical and surgical hospitals (M/S) and one specialty hospital (S).

Let us select the 11th row of the random numbers table: Note: Choosing a different row will result in a different sample
82173 26921 28472 98958 07960 66124 89731 95069

We select the first 2-digit number. If the number is between 00 and 23, then we select the corresponding location. If the number is greater than 23, then we ignore the number and select the next 2-digit number.

Repeat until 4 medical and surgical hospitals (M/S) and one specialty hospital (S) were selected (while any additional locations will also be ignored).

82	$\Rightarrow$	Ignore
17	$\Rightarrow$	Select St. Elizabeth Medical Center, North Unit (M/S)
32	$\Rightarrow$	Ignore
69	$\Rightarrow$	Ignore
21	$\Rightarrow$	Select Care Unit Hospital (S)
28	$\Rightarrow$	Ignore (we already selected one S)
47	$\Rightarrow$	Ignore
29	$\Rightarrow$	Ignore
89	$\Rightarrow$	Ignore
58	$\Rightarrow$	Ignore
07	$\Rightarrow$	Select Mercy Hospital-Anderson (M/S)
96	$\Rightarrow$	Ignore
06	$\Rightarrow$	Select Clermont Mercy Hospital (M/S)
61	$\Rightarrow$	Ignore
24	$\Rightarrow$	Ignore (we already selected one S)
89	$\Rightarrow$	Ignore
73	$\Rightarrow$	Ignore
19	$\Rightarrow$	Select St. Luke's Hospital West (M/S)

Thus the sample will then contain the five hospitals St. Elizabeth Medical Center North Unit, Care Unit Hospital, Mercy Hospital-Anderson, Clermont Mercy Hospital, St. Luke's Hospital West.

🎯 Result

(a) Children's Hospital Medical Center, St. Francis-St. George Hospital, Bethesda North, Good Samaritan Hospital, Mercy Hospital-Hamilton

(b) Answers could vary

(c) Jewish Hospital-Kenwood, Mercy Hospital-Anderson, Good Samaritan Hospital, St. Elizabeth Medical Center North Unit, Emerson Behavioral Science, Shriners Burns Institute.

(d) Answers could vary

PROBLEM STATEMENT

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Step 1

(a) Selected random numbers: 05, 20, 59, 21, 31, 28, 49, 38, 66, 08, 29, 02

The given ID numbers range from 00 to 34, thus we need to eliminate all random numbers that exceed 34:

$$05, 20, 21, 31, 28, 08, 29, 02$$

We only require five dealers, thus we will only use the first five ID numbers.

$$05, 20, 21, 31, 28$$

The corresponding dealers are then in the sample: Bob Schmidt Chevrolet, Great Lakes For Nissan, Grogan Towne Chrysler, Southside Lincoln Mercury, Rouen Chrysler Jeep Eagle.

Step 2

(b) Let us select the 11th row of the random numbers table: Note: Choosing a different row will result in a different sample

82173 26921 28472 98958 07960 66124

We select the first 2-digit number. If the number is between 00 and 34, then we select the corresponding location. If the number is greater than 34, then we ignore the number and select the next 2-digit number.

Repeat until 4 unique locations were selected.

82	$\Rightarrow$	Ignore
17	$\Rightarrow$	Select Doyle Pontiac Buick
32	$\Rightarrow$	Select Valiton Chrysler
69	$\Rightarrow$	Ignore
21	$\Rightarrow$	Select Grogan Towne Chrysler
28	$\Rightarrow$	Select Rouen Chrysler Jeep Eagle
47	$\Rightarrow$	Ignore
29	$\Rightarrow$	Select Saturn of Toledo

Thus the sample will then contain the five dealers: Doyle Pontiac Buick, Valiton Chrysler, Grogan Towne Chrysler, Rouen Chrysler Jeep Eagle, Saturn of Toledo.

Step 3

(c) We need to select every seventh location with starting point 04.

Let us first determine the Id numbers that will be in the sample, which are the starting point and the starting point increased by 7 or a multiple of 7.

The given ID numbers range from 00 to 34, thus we need to eliminate all random numbers that exceed 34.

$$\begin{aligned} & 04 \\ 04+07= & 11 \\ 11+07= & 18 \\ 18+07= & 25 \\ 25+07= & 32 \\ 32+07= & 39 \end{aligned}$$

The selected random numbers are then 04, 11, 18, 25, and 32. The corresponding dealers are: Yark Automotive Group, Thayer Chevrolet/Toyota, Franklin Park Lincoln Mercury, Mathews Ford Oregon Inc., Valiton Chrysler.

🎯 Result

(a) Bob Schmidt Chevrolet, Great Lakes For Nissan, Grogan Towne Chrysler, Southside Lincoln Mercury, Rouen Chrysler Jeep Eagle

(b) Answers could vary

(c) Yark Automotive Group, Thayer Chevrolet/Toyota, Franklin Park Lincoln Mercury, Mathews Ford Oregon Inc., Valiton Chrysler

PROBLEM STATEMENT

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Step 1

(a) Selected random numbers: 02, 59, 51, 25, 14, 29, 77, 69, 18

The given ID numbers range from 00 to 26, thus we need to eliminate all random numbers that exceed 26:

$$02, 25, 14, 18$$

The corresponding agents are then in the sample: Denker Brett, Wood Tom, Keisser Keith, Priest Harvey.

Step 2

(b) Let us select the 11th row of the random numbers table: Note: Choosing a different row will result in a different sample

82173 26921 28472 98958 07960 66124

We select the first 2-digit number. If the number is between 00 and 26, then we select the corresponding location. If the number is greater than 26, then we ignore the number and select the next 2-digit number.

Repeat until 4 unique locations were selected.

82	$\Rightarrow$	Ignore
17	$\Rightarrow$	Select O'Donnell Jim
32	$\Rightarrow$	Ignore
69	$\Rightarrow$	Ignore
21	$\Rightarrow$	Select Seibert John H
28	$\Rightarrow$	Ignore
47	$\Rightarrow$	Ignore
29	$\Rightarrow$	Ignore
89	$\Rightarrow$	Ignore
58	$\Rightarrow$	Ignore
07	$\Rightarrow$	Select Glemser Cathy
96	$\Rightarrow$	Ignore
06	$\Rightarrow$	Select Gibellato Carlo

Thus the sample will then contain the four agents: O'Donnell Jim, Seibert John H, Glemser Cathy, Gibellato Carlo.

Step 3

(c) We need to select every seventh location with starting point 04.

Let us first determine the Id numbers that will be in the sample, which are the starting point and the starting point increased by 7 or a multiple of 7.

The given ID numbers range from 00 to 26, thus we need to eliminate all random numbers that exceed 26.

$$\begin{aligned} & 04 \\ 04+07= & 11 \\ 11+07= & 18 \\ 18+07= & 25 \\ 25+07= & 32 \end{aligned}$$

The selected random numbers are then 04, 11, 18, and 25. The corresponding locations are: Farley Ron, Hinckley Dave, Priest Harvey, Wood Tom.

Result

(a) Denker Brett, Wood Tom, Keisser Keith, Priest Harvey

(b) Answers could vary

(c) Farley Ron, Hinckley Dave, Priest Harvey, Wood Tom

PROBLEM STATEMENT

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Step 1

Population values are : $12, 12, 14$ and 16.

a.

Total observations in the population is $N = 4$ and the sample size is $n = 2$ Total number of possible samples of size 2 is

$$\begin{aligned} {}_N C_n &= \frac{N!}{n!(N-n)!} \\ &= \frac{4!}{2!(4-2)!} \\ &= \frac{4!}{2!2!} \\ &= 6 \end{aligned}$$

Let X_i be the ith sample of size 2. So,

$$\bar{X}_i = \frac{X_{i1} + X_{i2}}{2}$$

Sample means for all possible samples of size 2 is given in the below table.

Step 2

Sample	Values	Sum $\sum_j X_{ij}$	Mean $\bar{X}_i$
1	12, 12	24	12
2	12, 14	26	13
3	12, 16	28	14
4	12, 14	26	13
5	12, 16	28	14
6	14, 16	30	15

b. Population mean is μ.

$$\begin{aligned}\mu &= \frac{12+12+14+16}{4}\\ &= \frac{54}{4}\\ &= 13.5\end{aligned}$$

Sampling distribution of the sample mean for $n = 2$ is as follows.

Sample Mean	Number of means	Probability
12	1	0.167
13	2	0.333
14	2	0.333
15	1	0.167

The sample mean is $\mu_{\bar{X}}$

$$\begin{aligned}\mu_{\bar{X}} &= \frac{\text{sum of all sample means}}{\text{total number of samples}}\\ &= \frac{12\times 1+13\times 2+14\times 2+15\times 1}{6}\\ &= \frac{12+26+28+15}{6}\\ &= \frac{81}{6}\\ &= 13.5\end{aligned}$$

This shows that $\mu = \mu_{\bar{X}}$.

Step 3

c. Population data is more dispersed than the sample data. The population observations vary from 12 to 16 but the sample mean observations vary from 12 to 16.

🎯 **Result**

a. 6 samples.

b. $\mu = 13.5$ and $\mu_{\bar{X}} = 13.5$

PROBLEM STATEMENT

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Step 1

Population values are : $2, 2, 4, 4$ and 8.

a.

Total observations in the population is $N = 5$ and the sample size is $n = 2$ Total number of possible samples of size 2 is

$$\begin{aligned} {}_N C_n &= \frac{N!}{n!(N-n)!} \\ &= \frac{5!}{2!(5-2)!} \\ &= \frac{5!}{2!3!} \\ &= 10 \end{aligned}$$

Let X_i be the ith sample of size 2. So,

$$\bar{X}_i = \frac{X_{i1} + X_{i2}}{2}$$

Sample means for all possible samples of size 2 is given in the below table.

Step 2

Sample	Values	Sum $\sum_j X_{ij}$	Mean $\bar{X}_i$
1	$2, 2$	4	2
2	$2, 4$	6	3
3	$2, 4$	6	3
4	$2, 8$	10	5
5	$2, 4$	6	3
6	$2, 4$	6	3
7	$2, 8$	10	5
8	$4, 4$	8	4
9	$4, 8$	12	6
10	$4, 8$	12	6

b. Population mean is μ.

$$\begin{aligned}\mu &= \frac{2+2+4+4+8}{5}\\ &= \frac{20}{5}\\ &= 4\end{aligned}$$

Sampling distribution of the sample mean for $n = 2$ is as follows.

Sample Mean	Number of means	Probability
2	1	0.1
3	4	0.4
4	1	0.1
5	2	0.2
6	2	0.2

The sample mean is $\mu_{\bar{X}}$

$$\begin{aligned}\mu_{\bar{X}} &= \frac{\text{sum of all sample means}}{\text{total number of samples}}\\ &= \frac{2\times 1+3\times 4+4\times 1+5\times 2+6\times 2}{10}\\ &= \frac{2+12+4+10+12}{10}\\ &= \frac{40}{10}\\ &= 4\end{aligned}$$

This shows that $\mu = \mu_{\bar{X}}$.

Step 3

c. Population data is more dispersed than the sample data. The population observations vary from 2 to 8 but the sample mean observations vary from 1 to 4.

🎯 **Result**

a. 10 samples

b. $\mu = 4$ and $\mu_{\bar{X}} = 4$

PROBLEM STATEMENT

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Step 1

Population values are : $12, 12, 14, 15$ and 20.

a.

Total observations in the population is $N = 5$ and the sample size is $n = 3$ Total number of possible samples of size 3 is

$$\begin{aligned} {}_N C_n &= \frac{N!}{n!(N-n)!} \\ &= \frac{5!}{3!(5-3)!} \\ &= \frac{5!}{3!2!} \\ &= 10 \end{aligned}$$

Let X_i be the ith sample of size 3. So,

$$\bar{X}_i = \frac{X_{i1} + X_{i2} + X_{i3}}{3}$$

Sample means for all possible samples of size 3 is given in the below table.

Step 2

Sample	Values	Sum $\sum_j X_{ij}$	Mean $\bar{X}_i$
1	12, 12, 14	38	38/3
2	12, 12, 15	39	39/3
3	12, 12, 20	44	44/3
4	12, 14, 15	41	41/3
5	12, 14, 20	46	46/3
6	12, 15, 20	47	47/3
7	12, 14, 15	41	41/3
8	12, 14, 20	46	46/3
9	12, 15, 20	47	47/3
10	14, 15, 20	49	49/3

b. Population mean is μ.

$$\begin{aligned}\mu &= \frac{12+12+14+15+20}{5}\\ &= \frac{73}{5}\\ &= 14.6\end{aligned}$$

Sampling distribution of the sample mean for $n = 3$ is as follows.

Sample Mean	Number of means	Probability
38/3	1	0.1
39/3	1	0.1
41/3	2	0.2
44/3	1	0.1
46/3	2	0.2
47/3	2	0.2
49/3	1	0.1

Step 3

The sample mean is $\mu_{\bar{X}}$

$$\begin{aligned}\mu_{\bar{X}} &= \frac{\text{sum of all sample means}}{\text{total number of samples}} \\ &= \frac{\frac{38}{3}\times 1+\frac{39}{3}\times 1+\frac{41}{3}\times 2+\frac{44}{3}\times 1+\frac{46}{3}\times 2+\frac{47}{3}\times 2+\frac{49}{3}\times 1}{10} \\ &= \frac{\frac{38+39+82+44+92+94+49}{3}}{10} \\ &= \frac{\frac{438}{3}}{10} \\ &= \frac{146}{10} \\ &= 14.6\end{aligned}$$

This shows that $\mu = \mu_{\bar{X}}$.

c. Population data is more dispersed than the sample data. The population observations vary from 12 to 20 but the sample mean observations vary from 12.67 to 16.33.

Result

a. 10 samples

b. $\mu = 14.6$ and $\mu_{\bar{X}} = 14.6$

PROBLEM STATEMENT

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Step 1

Population values are : $0, 0, 1, 3$ and 6.

a.

Total observations in the population is $N = 5$ and the sample size is $n = 3$ Total number of possible samples of size 2 is

$$\begin{aligned} {}_N C_n &= \frac{N!}{n!(N-n)!} \\ &= \frac{5!}{2!(5-2)!} \\ &= \frac{5!}{2!3!} \\ &= 10 \end{aligned}$$

Let X_i be the ith sample of size 3. So,

$$\bar{X}_i = \frac{X_{i1} + X_{i2} + X_{i3}}{3}$$

Sample means for all possible samples of size 3 is given in the below table.

Step 2

Sample	Values	Sum $\sum_j X_{ij}$	Mean $\bar{X}_i$
1	$0, 0, 1$	1	$1/3$
2	$0, 0, 3$	3	$3/3$
3	$0, 0, 6$	6	$6/3$
4	$0, 1, 3$	4	$4/3$
5	$0, 1, 6$	7	$7/3$
6	$0, 3, 6$	9	$9/3$
7	$0, 1, 3$	4	$4/3$
8	$0, 1, 6$	7	$7/3$
9	$0, 3, 6$	9	$9/3$
10	$1, 3, 6$	10	$10/3$

b. Population mean is μ.

$$\begin{aligned}\mu &= \frac{0+0+1+3+6}{5} \\ &= \frac{10}{5} \\ &= 2\end{aligned}$$

Sampling distribution of the sample mean for $n = 3$ is as follows.

Sample Mean	Number of means	Probability
$1/3$	1	0.1
$3/3$	1	0.1
$4/3$	2	0.2
$6/3$	1	0.1
$7/3$	2	0.2
$9/3$	2	0.2
$10/3$	1	0.1

Step 3

The sample mean is $\mu_{\bar{X}}$

$$\begin{aligned}\mu_{\bar{X}} &= \frac{\text{sum of all sample means}}{\text{total number of samples}} \\ &= \frac{\frac{1}{3}\times 1+\frac{3}{3}\times 1+\frac{4}{3}\times 2+\frac{6}{3}\times 1+\frac{7}{3}\times 2+\frac{9}{3}\times 2+\frac{10}{3}\times 1}{10} \\ &= \frac{\frac{1+3+8+6+14+18+10}{3}}{10} \\ &= \frac{\frac{60}{3}}{10} \\ &= \frac{20}{10} \\ &= 2\end{aligned}$$

This shows that $\mu = \mu_{\bar{X}}$.

c. Population data is more dispersed than the sample data. The population observations vary from 0 to 6 but the sample mean observations vary from 0.33 to 3.33.

Result

a. 10 samples

b. $\mu = 2$ and $\mu_{\bar{X}} = 2$

PROBLEM STATEMENT

Chapter 8, Page 278

Step 1

Given:

Associate	Number of Cases
Ruud	3
Wu	6
Sass	3
Flores	3
Wilhelms	0
Schueller	1

(a) We need to determine the number of possible samples of size 3, which is the number of ways to select 3 of the 6 associates.

$$n = 6$$
$$r = 3$$

The order of the associates is not important, because a different order of the associates leads to the same sample. This then implies that we need to use the definition of a **combination**: ${}_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$

$${}_6C_3 = \frac{6!}{3!(6-3)!} = \frac{6!}{3!3!} = 20$$

Thus there are 20 different samples of size 3.

Step 2

(b) We need to list all possible samples of size 3 and determine the mean of the number of cases, which is the sum of all number of cases divided by 3 (as there are 3 elements in the sample).

Sample	Sample mean
Ruud, Wu, Sass	$(3+6+3)/3 = 4$
Ruud, Wu, Flores	$(3+6+3)/3 = 4$
Ruud, Wu, Wilhelms	$(3+6+0)/3 = 3$
Ruud, Wu, Schueller	$(3+6+1)/3 \approx 3.3333$
Ruud, Sass, Flores	$(3+3+3)/3 = 3$
Ruud, Sass, Wilhelms	$(3+3+0)/3 = 2$
Ruud, Sass, Schueller	$(3+3+1)/3 \approx 2.3333$
Ruud, Flores, Wilhelms	$(3+3+0)/3 = 2$
Ruud, Flores, Schueller	$(3+3+1)/3 \approx 2.3333$
Ruud, Wilhelms, Schueller	$(3+0+1)/3 \approx 1.3333$
Wu, Sass, Flores	$(6+3+3)/3 = 4$
Wu, Sass, Wilhelms	$(6+3+0)/3 = 3$
Wu, Sass, Schueller	$(6+3+1)/3 \approx 3.3333$
Wu, Flores, Wilhelms	$(6+3+0)/3 = 3$
Wu, Flores, Schueller	$(6+3+1)/3 \approx 3.3333$
Wu, Wilhelms, Schueller	$(6+0+1)/3 \approx 2.3333$
Sass, Flores, Wilhelms	$(3+3+0)/3 = 2$
Sass, Flores, Schueller	$(3+3+1)/3 \approx 2.3333$
Sass, Wilhelms, Schueller	$(3+0+1)/3 \approx 1.3333$
Flores, Wilhelms, Schueller	$(3+0+1)/3 \approx 1.3333$

Step 3

(c) Let us first determine the mean of all sample means in part (b):

$$\begin{aligned}
&\text{Mean of all sample means} \\
&= \frac{\sum x_i}{n} \\
&= \frac{\begin{array}{c} 4+4+3+3.3333+3 \\ +2+2.3333+2+2.3333+1.3333 \\ +4+3+3.3333+3+3.3333 \\ +2.3333+2+2.3333+1.3333+1.3333 \end{array}}{20} \\
&= \frac{53.333}{20} \\
&\approx 2.66665
\end{aligned}$$

Next, we determine the population mean, which is the sum of the number of cases of all associates divided by the number of associates.

$$\begin{aligned}
\text{Population mean} &= \frac{\sum x_i}{N} \\
&= \frac{3+6+3+3+0+1}{6} \\
&= \frac{16}{6} \\
&\approx 2.66667
\end{aligned}$$

We note that the mean of the distribution of sample means is equal to the population mean (the very small difference in values is due to rounding errors).

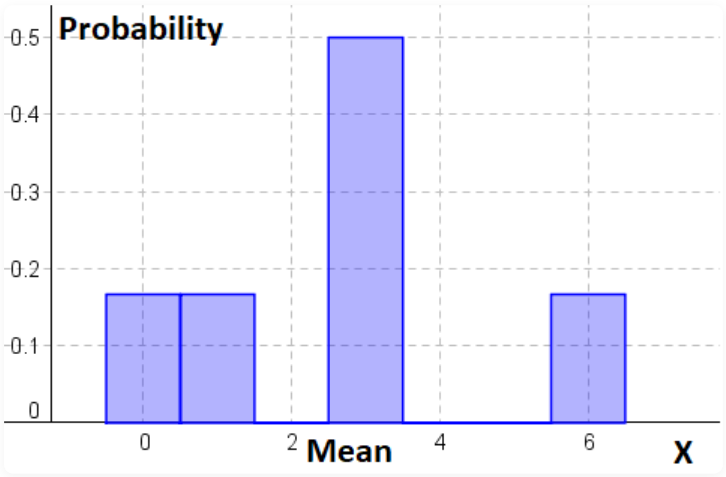
Step 4

(d) **Population distribution**

The probability is the number of data values with the specific value divided by the number of data values in total.

X	Probability
0	1/6
1	1/6
3	3/6
6	1/6

The bars need to have the same width. The height of the bars need to be equal to the probability.



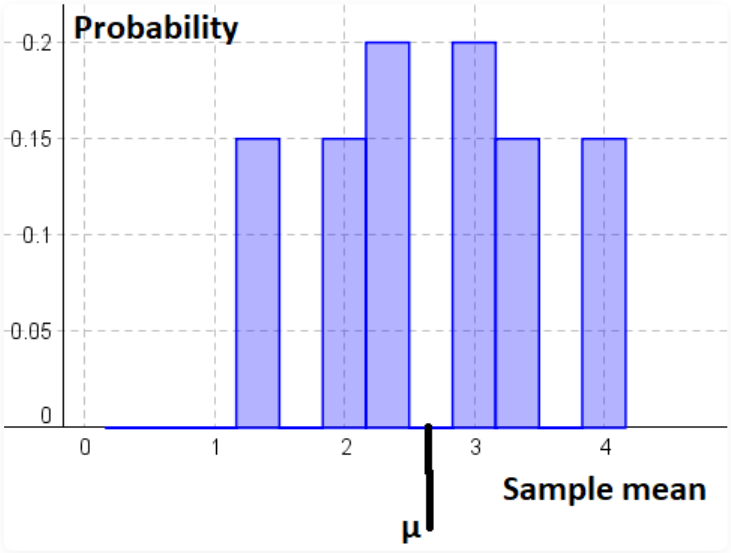
Step 5

Distribution sample mean

The probability is the number of data values with the specific value divided by the number of data values in total.

X	Probability
1.3333	3/20
2	3/20
2.3333	4/20
3	4/20
3.3333	3/20
4	3/20

The bars need to have the same width. The height of the bars need to be equal to the probability.



Result

(a) 20 different samples of size 3

(b) 4, 4, 3, 3.3333, 3, 2, 2.3333, 2, 2.3333, 1.3333, 4, 3, 3.3333, 3, 3.3333, 2.3333, 2, 2.3333, 1.3333, 1.3333

(c) The mean of the distribution of sample means is equal to the population mean.

(d) See graphs

PROBLEM STATEMENT

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Step 1

Given:

Representative	Cars Sold
Peter	8
Connie	6
Juan	4
Ted	10
Peggy	6

(a) We need to determine the number of possible samples of size 2, which is the number of ways to select 2 of the 5 representatives.

$$n = 5$$
$$r = 2$$

The order of the associates is not important, because a different order of the associates leads to the same sample. This then implies that we need to use the definition of a **combination**: ${}_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$

$${}_5C_2 = \frac{5!}{2!(5-2)!} = \frac{5!}{2!3!} = 10$$

Thus there are 10 different samples of size 2.

Step 2

(b) We need to list all possible samples of size 2 and determine the mean of the number of cases, which is the sum of all number of cases divided by 2 (as there are 2 elements in the sample).

Sample	Sample mean
Peter, Connie	$(8+6)/2 = 7$
Peter, Juan	$(8+4)/2 = 6$
Peter, Ted	$(8+10)/2 = 9$
Peter, Peggy	$(8+6)/2 = 7$
Connie, Juan	$(6+4)/2 = 5$
Connie, Ted	$(6+10)/2 = 8$
Connie, Peggy	$(6+6)/2 = 6$
Juan, Ted	$(4+10)/2 = 7$
Juan, Peggy	$(4+6)/2 = 5$
Ted, Peggy	$(10+6)/2 = 8$

Step 3

(c) Let us first determine the mean of all sample means in part (b):

$$\begin{aligned}&\text{Mean of all sample means}\\&=\frac{\sum x_i}{n}\\&=\frac{\begin{array}{r}7+6+9+7+5\\+8+6+7+5+8\end{array}}{10}\\&=\frac{68}{10}\\&=6.8\end{aligned}$$

Next, we determine the population mean, which is the sum of the number of cases of all associates divided by the number of associates.

$$\begin{aligned}\text{Population mean}&=\frac{\sum x_i}{N}\\&=\frac{8+6+4+10+6}{5}\\&=\frac{34}{5}\\&=6.8\end{aligned}$$

We note that the mean of the distribution of sample means is equal to the population mean.

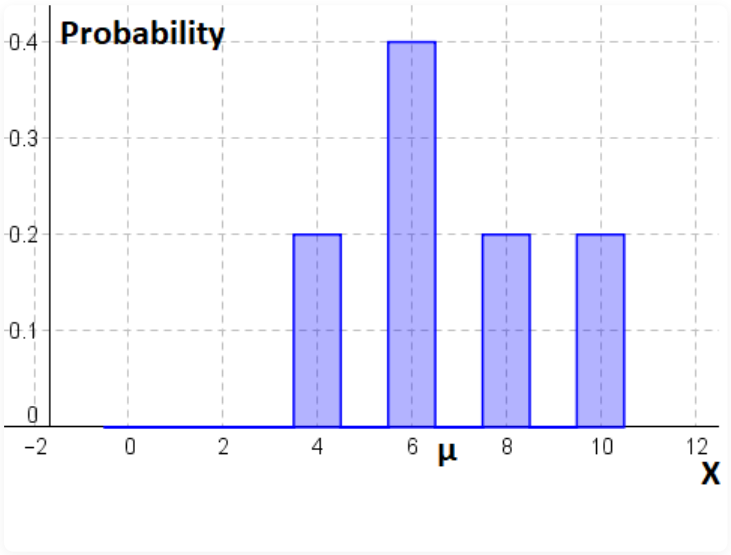
Step 4

(d) **Population distribution**

The probability is the number of data values with the specific value divided by the number of data values in total.

X	Probability
4	1/5
6	2/5
8	1/5
10	1/5

The bars need to have the same width. The height of the bars need to be equal to the probability.



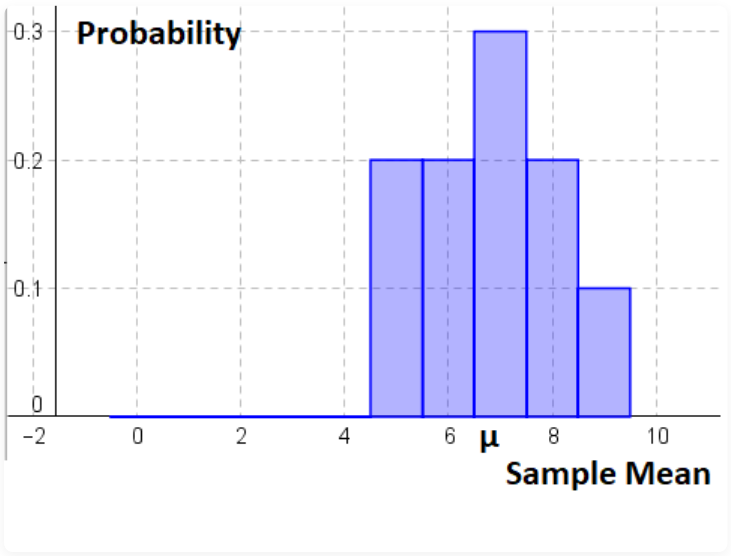
Step 5

Distribution sample mean

The probability is the number of data values with the specific value divided by the number of data values in total.

X	Probability
5	2/10
6	2/10
7	3/10
8	2/10
9	1/10

The bars need to have the same width. The height of the bars need to be equal to the probability.



Result

(a) 10 different samples of size 2

(b) 7, 6, 9, 7, 5, 8, 6, 7, 5, 8

(c) The mean of the distribution of sample means is equal to the population mean.

(d) See graphs

PROBLEM STATEMENT

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Step 1

(a) All digits from 0 to 9 are equally likely.

Since there are 10 possible digits from 0 to 9, we have 1 chance in 10 of selecting each digit.

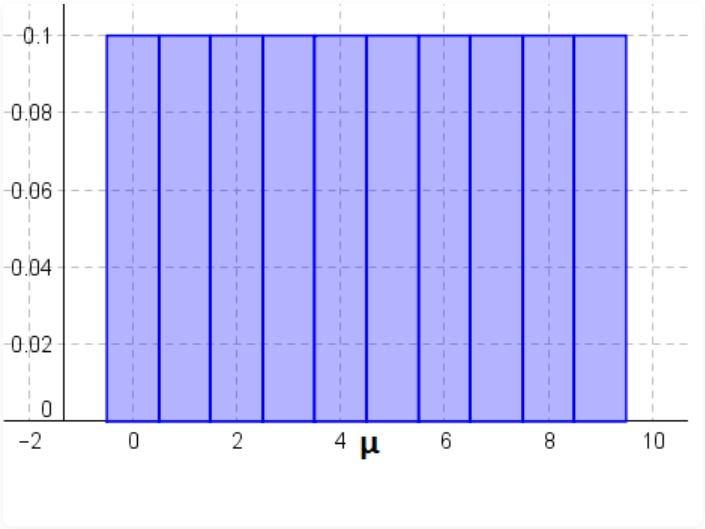
$$P(X = x) = \frac{1}{10} = 0.1$$

The population mean is the sum of all possible values (digits) divided by the number of values (digits).

$$\mu = \frac{\sum x_i}{n} = \frac{0+1+2+3+4+5+6+7+8+9}{10} = \frac{45}{10} = 4.5$$

Probability distribution

The width of the bars have to be the same, while the height of the bars need to be equal to the probability of 0.1.



Step 2

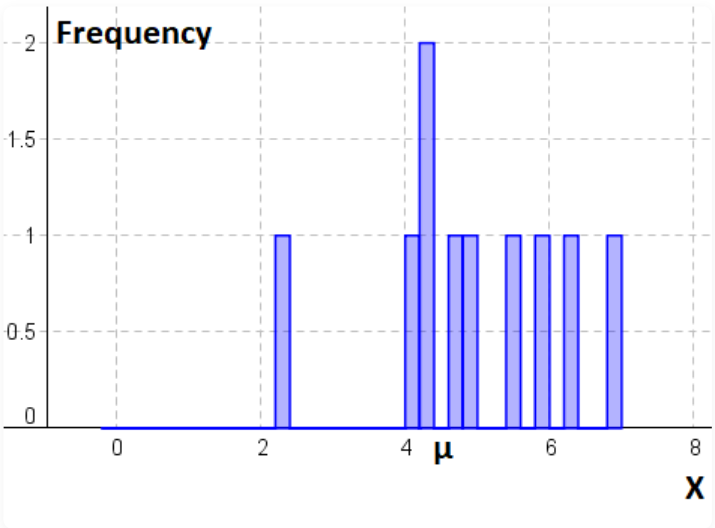
(b) We consider each row to be a separate sample. We need to determine the sample mean of each of the samples, which is obtained by adding the five digits and dividing by 5.

$$\overline{x} = (0+2+7+1+1)/5 = 2.2$$
$$\overline{x} = (9+4+8+7+3)/5 = 6.2$$
$$\overline{x} = (5+4+9+2+1)/5 = 4.2$$
$$\overline{x} = (7+7+6+4+0)/5 = 4.8$$
$$\overline{x} = (6+1+5+4+5)/5 = 4.2$$
$$\overline{x} = (1+7+1+4+7)/5 = 4$$
$$\overline{x} = (1+3+7+4+8)/5 = 4.6$$
$$\overline{x} = (8+7+4+5+5)/5 = 5.8$$
$$\overline{x} = (0+8+9+9+9)/5 = 7$$
$$\overline{x} = (7+8+8+0+4)/5 = 5.4$$

Next, we determine the mean of all sample means:

$$\begin{aligned}&\text{Mean of sample means}\\ &= \frac{\sum x_i}{n}\\ &= \frac{2.2+6.2+4.2+4.8+4.2+4+4.6+5.8+7+5.4}{10}\\ &= \frac{48.4}{10}\\ &= 4.8\end{aligned}$$

The mean of the sample distribution of the sample means (4.8) is slightly greater than the population mean (4.5), due to the samples not containing all possible samples of size 5.



🎯 Result

(a) 4.5

(b) 2.2, 6.2, 4.2, 4.8, 4.2, 4, 4.6, 5.8, 7, 5.4

PROBLEM STATEMENT

Chapter 8, Page 286

Step 1

Given population values: 2, 3, 2, 3, 3, 4, 2, 4, 3, 2, 2, 7, 3, 4, 5, 3, 3, 3, 3, 5

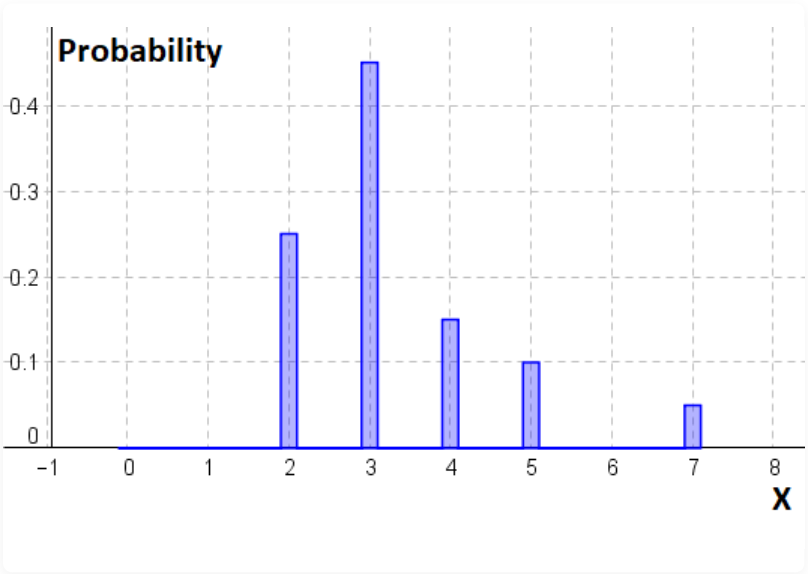
(a) The frequency is the number of times that the data value occurs.

The probability is the frequency divided by the total number of population values (which is 20).

Value	Frequency	Probability
2	5	$5/20 = 0.25$
3	9	$9/20 = 0.45$
4	3	$3/20 = 0.15$
5	2	$2/20 = 0.10$
7	1	$1/20 = 0.05$

Graph

The bars need to have the same width, while the height is equal to the probability.



Step 2

(b) The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned}\mu &= \frac{\sum_{i=1}^{n} x_i}{n} \\ &= \frac{\begin{array}{c}2+3+2+3+3\\+4+2+4+3+2\\+2+7+3+4+5\\+3+3+3+3+5\end{array}}{20} \\ &= \frac{66}{20} \\ &= 3.3\end{aligned}$$

Step 3

(c) We need to select five random samples of size 5 each.

We will use the first rows of the table to select these five random samples. We select the first five 2-digit numbers in the rows that are between 01 and 20, while we then add the corresponding data value to the sample. *Note: Using different rows of the table will result in different samples.*

First sample: 2, 2, 3, 3, 2

Second sample: 3, 4, 3, 4, 3

Third sample: 2, 3, 3, 3, 3

Fourth sample: 3, 4, 3, 3, 4

Fifth sample: 2, 3, 4, 3, 3

Next, we need to compute the sample mean of each of these samples, which is the sum of data values divided by the number of data values.

$$\begin{aligned}\overline{x} &= (2+2+3+3+2)/5 = 2.4 \\ \overline{x} &= (3+4+3+4+3)/5 = 3.4 \\ \overline{x} &= (2+3+3+3+3)/5 = 2.8 \\ \overline{x} &= (3+4+3+3+4)/5 = 3.4 \\ \overline{x} &= (2+3+4+3+3)/5 = 3\end{aligned}$$

Step 4

(d) We need to determine the mean of all sample means in part (c):

$$\begin{aligned}\text{Mean of sample means} &= \frac{2.4+3.4+2.8+3.4+3}{5}\\ &= \frac{15}{5}\\ &= 3\end{aligned}$$

We note that the mean of the sample means (3) is lower than the population mean (3).

We would not expect the mean of the sample means to be equal to the population mean, because the samples in part (c) are not all possible samples of size 5 while the mean of all possible sample means will be equal to the population mean.

Step 5

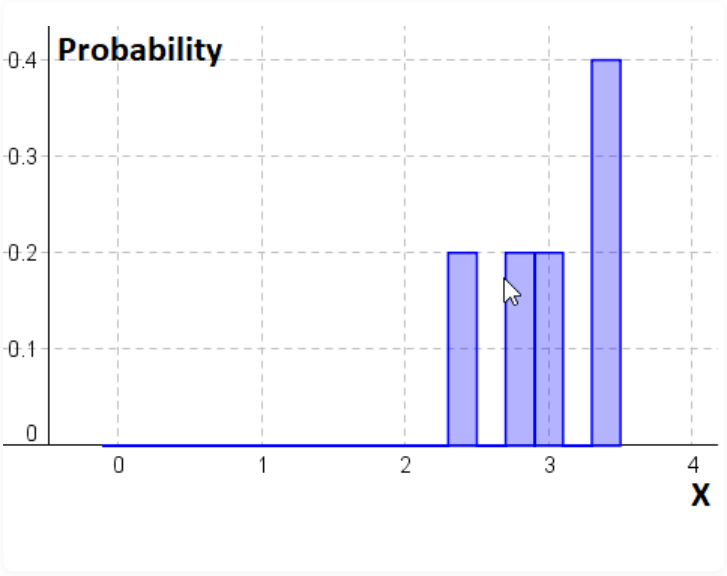
(e) The frequency is the number of times that the data value occurs.

The probability is the frequency divided by the total number of population values (which is 5).

Sample mean	Frequency	Probability
2.4	1	$1/5 = 0.2$
2.8	1	$1/5 = 0.2$
3	1	$1/5 = 0.2$
3.4	2	$2/5 = 0.4$

We note that the probability distribution in part (a) is positively skewed, because the highest bars are to the left in the histogram.

However, the sample means are negatively skewed, because the highest bars are to the right in the histogram.



🎯 Result

(a)

Value	Frequency	Probability
2	5	0.25
3	9	0.45
4	3	0.15
5	2	0.10
7	1	0.05

(b) 3.3

(c) Answers will vary

(d) No

(e) Yes

PROBLEM STATEMENT

Chapter 8, Page 286

Step 1

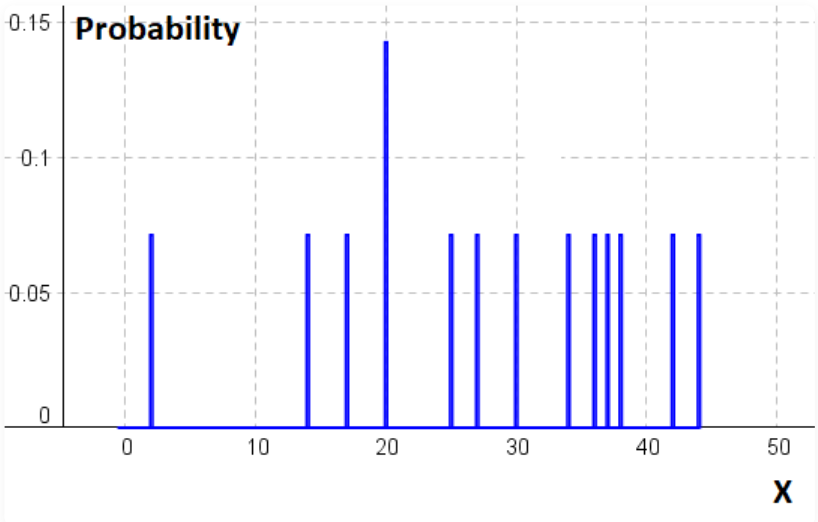
(a) For example, the years mentioned on the coins in my purse are: 1976, 1986, 2018, 1986, 1978, 1983, 2000, 2000, 2003, 2006, 1982, 1993, 1995, 1990.

This solution was written in 2020, thus the ages of the coins are then: 44, 36, 2, 34, 42, 37, 20, 20, 17, 14, 38, 27, 25, 30.

The **frequency** is the number of times a certain data value occur.

The **probability** is the frequency divided by the total number of data values (14 in this case).

Value	Frequency	Probability
2	1	$1/14 \approx 0.0714$
14	1	$1/14 \approx 0.0714$
17	1	$1/14 \approx 0.0714$
20	2	$2/14 \approx 0.1428$
25	1	$1/14 \approx 0.0714$
27	1	$1/14 \approx 0.0714$
30	1	$1/14 \approx 0.0714$
34	1	$1/14 \approx 0.0714$
36	1	$1/14 \approx 0.0714$
37	1	$1/14 \approx 0.0714$
38	1	$1/14 \approx 0.0714$
42	1	$1/14 \approx 0.0714$
44	1	$1/14 \approx 0.0714$



Step 2

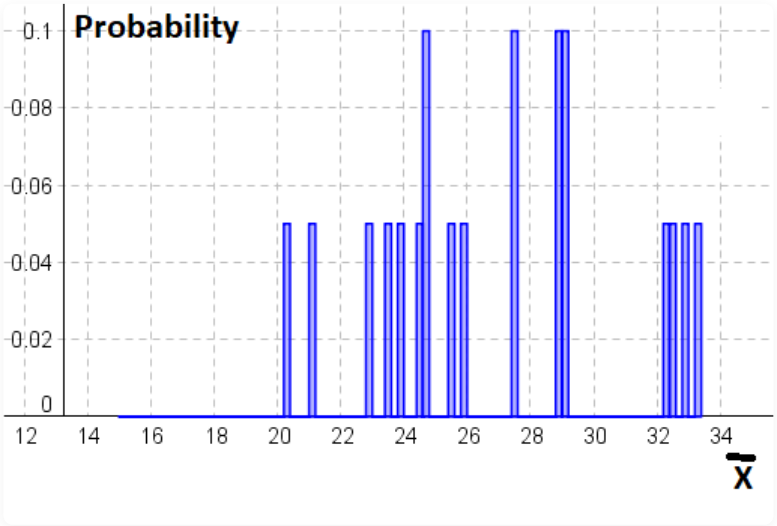
(b) We need to select five random samples of size 5 each.

We will use the rows of the random digits table to select these five random samples. We select the first five 2-digit numbers in the rows that are between 01 and 14, while we then add the corresponding data value to the sample. Note: Using different rows of the table will result in different samples.

First sample: 44, 20, 38, 25, 34
Second sample: 2, 14, 25, 44, 20
Third sample: 38, 30, 20, 14, 20
....

Next, we need to compute the sample mean of each of these samples, which is the sum of data values divided by the number of data values.

$$\overline{x} = (44 + 20 + 38 + 25 + 34)/5 = 32.2$$
$$\overline{x} = (2 + 14 + 25 + 44 + 20)/5 = 21$$
$$\overline{x} = (38 + 30 + 20 + 14 + 20)/5 = 24.4$$
....



Step 3

(c) We note that the distribution of the sample means is symmetric and approximately bell-shaped, because the highest bars are in the middle of the histogram.

The distribution of the probability distribution is also roughly symmetric, because the highest bars are roughly in the middle of the histogram. However, the probability distribution appears to be wider than the distribution of the sample means.

🎯 Result

Answers will vary

PROBLEM STATEMENT

Chapter 8, Page 286

Step 1

Note: Answers will vary when you use different samples.

(a) For example, I will use the first 30 digits in the random digit table as example of the final digit of the selected phone numbers.

0, 2, 7, 1, 1, 0, 8, 1, 8, 2, 7, 5, 9, 9, 7, 7, 9, 8, 6, 6, 5, 8, 0, 9, 5, 8, 3, 3, 1, 9

The **frequency** is the number of times a certain data value occur.

Value	**Frequency**
0	3
1	4
2	2
3	2
4	0
5	3
6	2
7	4
8	5
9	5

Since digits can take on the values from 0 to 9, the probability distribution is approximately a uniform distributions with boundaries 0 and 9.

The mean of a uniform distribution is the average of the boundaries:

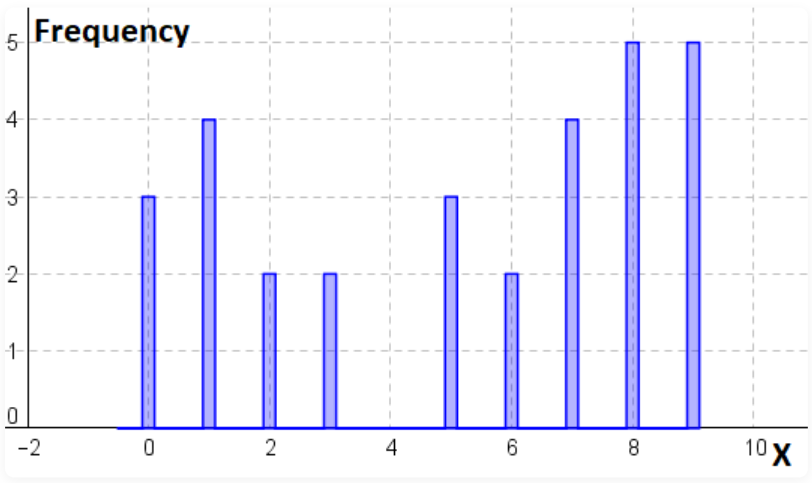
$$\mu = \frac{\alpha + \beta}{2} = \frac{0 + 9}{2} = \frac{9}{2} = 4.5$$

The variance of a uniform distribution is the squared difference of the boundaries divided by the 12:

$$\sigma^2 = \frac{(\beta - \alpha)^2}{12} = \frac{(9 - 0)^2}{12} = \frac{81}{12} = 6.75$$

The standard deviation is the square root of the variance:

$$\sigma = \sqrt{\sigma^2} = \sqrt{6.75} \approx 2.5981$$



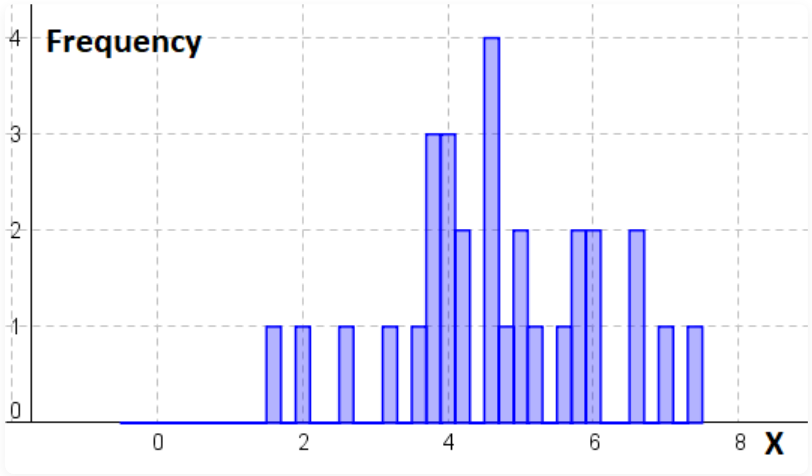
Step 2

(b) We will use the first four digits of the first 30 rows in the random digit table as examples of the final four digits of 30 randomly selected phone numbers.

Next, we need to compute the sample mean of each of these samples, which is the sum of data values divided by the number of data values. We then obtain the following sample means:

$$2.5, 7, 5, 6, 4, 3.25, 3.75, 6, 6.5, 5.75,$$
$$4.5, 7.5, 6.5, 4.25, 5.25, 4.5, 4.5, 4, 5.75, 4,$$
$$4.75, 5.5, 3.75, 3.5, 5, 4.5, 1.5, 3.75, 2, 4.25$$

The **frequency** is the number of times a certain data value occur.



Step 3

(c) We note that the distribution of the sample means is symmetric and approximately bell-shaped, because the highest bars are in the middle of the histogram.

The distribution of the probability distribution is also roughly symmetric but bimodal, because the highest bars are on the far left and on the far right side of the graph.

Thus the two graphs do not have approximately the same shape

Result

(a) Mean 6.75 and standard deviation 2.5981

(b)-(c) Answers could vary

PROBLEM STATEMENT

Chapter 8, Page 289

Step 1

Mean of the normal distribution is $\mu = 60$ and the standard deviation is $\sigma = 12$. Sample size is $n = 9$.

Let $\bar{X}$ be the sample mean.

$\textbf{a. } \bar{X}$ is greater than 63 }$

$$\begin{aligned} P(\bar{X} > 63) &= P(\bar{X} - \mu > 63 - \mu) \\ &= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{63 - \mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(z > \frac{63 - 60}{12/\sqrt{9}}\right) \\ &= P\left(z > \frac{3}{4}\right) \\ &= P(z > 0.75) \\ &= 0.5 - P(0 < z < 0.75) \\ &= 0.5 - 0.2734 \ \ (\text{from Appendix B1.}) \\ &= 0.2266 \end{aligned}$$

Step 2

$\textbf{b. } \bar{X}$ is less than 56 }$

$$\begin{aligned} P(\bar{X} < 56) &= P(\bar{X} - \mu < 56 - \mu) \\ &= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{56 - \mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(z < \frac{56 - 60}{12/\sqrt{9}}\right) \\ &= P\left(z < \frac{-4}{4}\right) \\ &= P(z < -1) \\ &= P(z > 1) \\ &= 0.5 - P(0 < z < 1) \\ &= 0.5 - 0.3413 \ \ (\text{from Appendix B1.}) \\ &= 0.1587 \end{aligned}$$

Step 3

$\textbf{c. } \bar{X}$ is between 56 and 63 }$

$$\begin{aligned} P(56 < \bar{X} < 63) &= P(56 < \bar{X} - \mu < 63 - \mu) \\ &= P\left(\frac{56 - \mu}{\sigma/\sqrt{n}} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{63 - \mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(\frac{56 - 60}{12/\sqrt{9}} < z < \frac{63 - 60}{12/\sqrt{9}}\right) \\ &= P\left(\frac{-4}{4} < z < \frac{3}{4}\right) \\ &= P(-1 < z < 0.75) \end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-1 < z < 0), P(0 < z < 0.75)$ and add the two probabilities.

$$\begin{aligned} P(56 < \bar{X} < 63) &= P(-1 < z < 0) + P(0 < z < 0.75) \\ &= P(0 < z < 1) + P(0 < z < 0.75) \\ &= 0.3413 + 0.2734 \ \ (\text{from Appendix B1.}) \\ &= 0.6147 \end{aligned}$$

Result

a. 0.2266

b. 0.1587

Chapter 8 Exercise 16

Page 263

Step 1

4. Step & Result Cards Not found

PROBLEM STATEMENT

Chapter 8, Page 289

Step 1

Mean of the normal distribution is $\mu = 75$ and the standard deviation is $\sigma = 5$. Sample size is $n = 40$.

Let $\bar{X}$ be the sample mean.

$\textbf{a. $\bar{X}$ is less than 74 }$

$$\begin{aligned} P(\bar{X} < 74) &= P(\bar{X} - \mu < 74 - \mu) \\ &= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{74 - \mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(z < \frac{74 - 75}{5/\sqrt{40}}\right) \\ &= P\left(z < \frac{-1}{0.79}\right) \\ &= P(z < -1.26) \\ &= P(z > 1.26) \\ &= 0.5 - P(0 < z < 1.26) \\ &= 0.5 - 0.3962 \ \ (\text{from Appendix B1.}) \\ &= 0.1038 \end{aligned}$$

Step 2

$\textbf{b. } \bar{X}$ is between 74 and 76 }$

$$\begin{aligned} P(74 < \bar{X} < 76) &= P(74 < \bar{X} - \mu < 76 - \mu) \\ &= P\left(\frac{74-\mu}{\sigma/\sqrt{n}} < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < \frac{76-\mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(\frac{74-75}{5/\sqrt{40}} < z < \frac{76-75}{5/\sqrt{40}}\right) \\ &= P\left(\frac{-1}{0.79} < z < \frac{1}{0.79}\right) \\ &= P(-1.26 < z < 1.26) \end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-1.26 < z < 0)$, $P(0 < z < 1.26)$ and add the two probabilities.

$$\begin{aligned} P(74 < \bar{X} < 76) &= P(-1.26 < z < 0) + P(0 < z < 1.26) \\ &= P(0 < z < 1.26) + P(0 < z < 1.26) \\ &= 0.3962 + 0.3962 \ \ (\text{from Appendix B1.}) \\ &= 0.7924 \end{aligned}$$

Step 3

$\textbf{c. } \bar{X}$ is between 76 and 77 }$

$$\begin{aligned} P(76 < \bar{X} < 77) &= P(76 < \bar{X} - \mu < 77 - \mu) \\ &= P\left(\frac{76-\mu}{\sigma/\sqrt{n}} < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < \frac{77-\mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(\frac{76-75}{5/\sqrt{40}} < z < \frac{77-75}{5/\sqrt{40}}\right) \\ &= P\left(\frac{1}{0.79} < z < \frac{2}{0.79}\right) \\ &= P(1.26 < z < 2.53) \end{aligned}$$

The two points are on the same sides of the mean, so calculate $P(0 < z < 1.26)$, $P(0 < z < 2.53)$ and add the two probabilities.

$$\begin{aligned} P(76 < \bar{X} < 77) &= P(0 < z < 2.53) - P(0 < z < 1.26) \\ &= 0.4943 - 0.3962 \ \ (\text{from Appendix B1.}) \\ &= 0.0981 \end{aligned}$$

Step 4

$\textbf{a. } \bar{X}$ is greater than 77 }$

$$
\begin{aligned}
P(\bar{X} > 77) &= P(\bar{X} - \mu > 77 - \mu) \\
&= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{77 - \mu}{\sigma/\sqrt{n}}\right) \\
&= P\left(z > \frac{77 - 75}{5/\sqrt{40}}\right) \\
&= P\left(z > \frac{2}{0.79}\right) \\
&= P(z > 2.53) \\
&= 0.5 - P(0 < z < 2.53) \\
&= 0.5 - 0.4943 \ \ (\text{from Appendix B1.}) \\
&= 0.0057
\end{aligned}
$$

Result

a. 0.1038

b. 0.7924

c. 0.0981

d. 0.0057

PROBLEM STATEMENT

Chapter 8, Page 289

Step 1

Mean rent for a one-bedroom apartment is $\mu = \$2200$ per month and the standard deviation is $\sigma = \$250$ per month. Sample size is $n = 50$.
Let $\bar{X}$ be the sample mean of one-bedroom apartment.

$\bar{X}$ is greater than 1950

$$
\begin{aligned}
P(\bar{X} > 1950) &= P(\bar{X} - \mu > 1950 - \mu) \\
&= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{1950 - \mu}{\sigma/\sqrt{n}}\right) \\
&= P\left(z > \frac{1950 - 2200}{250/\sqrt{50}}\right) \\
&= P\left(z > \frac{-250}{35.36}\right) \\
&= P(z > -7.07) \\
&= P(z < 7.07) \\
&= 0.5 + P(0 < z < 7.07) \\
&= 0.5 + 0.5 \quad \text{Probability is for any } z \text{ value greater than 3 is 0.5} \\
&= 1
\end{aligned}
$$

Result

1

PROBLEM STATEMENT

Chapter 8, Page 289

Step 1

Mean time taken by taxpayers is $\mu = 330$ minutes and the standard deviation is $\sigma = 80$ minutes. Sample size of taxpayers is $n = 40$.

Let $\bar{X}$ be the sample mean.

a. Standard error of the mean

$$\begin{aligned}\sigma/\sqrt{n} &= 80/\sqrt{40} \\ &= 80/6.32 \\ &= 12.66\end{aligned}$$

$\textbf{b. $\bar{X}$ is greater than 320 }$

$$\begin{aligned}P(\bar{X} > 320) &= P(\bar{X} - \mu > 320 - \mu) \\ &= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{320 - \mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(z > \frac{320 - 330}{80/\sqrt{40}}\right) \\ &= P\left(z > \frac{-10}{12.66}\right) \\ &= P(z > -0.79) \\ &= P(z < 0.79) \\ &= 0.5 + P(0 < z < 0.79) \\ &= 0.5 + 0.2852 \ \ (\text{from Appendix B1.}) \\ &= 0.7852\end{aligned}$$

Step 2

$\textbf{c. } \bar{X}$ is between 320 and 350 }$

$$\begin{aligned} P(320 < \bar{X} < 350) &= P(320 < \bar{X} - \mu < 350 - \mu) \\ &= P\left(\frac{320 - \mu}{\sigma/\sqrt{n}} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{350 - \mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(\frac{320 - 330}{80/\sqrt{40}} < z < \frac{350 - 330}{80/\sqrt{40}}\right) \\ &= P\left(\frac{-10}{12.66} < z < \frac{20}{12.66}\right) \\ &= P(-0.79 < z < 1.58) \end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-0.79 < z < 0)$, $P(0 < z < 1.58)$ and add the two probabilities.

$$\begin{aligned} P(320 < \bar{X} < 350) &= P(-0.79 < z < 0) + P(0 < z < 1.58) \\ &= P(0 < z < 0.79) + P(0 < z < 1.58) \\ &= 0.2852 + 0.4429 \ \ (\text{from Appendix B1.}) \\ &= 0.7281 \end{aligned}$$

Step 3

$\textbf{a. } \bar{X}$ is greater than 350 }$

$$\begin{aligned} P(\bar{X} > 350) &= P(\bar{X} - \mu > 350 - \mu) \\ &= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{350 - \mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(z > \frac{350 - 330}{80/\sqrt{40}}\right) \\ &= P\left(z > \frac{20}{12.66}\right) \\ &= P(z > 1.58) \\ &= 0.5 - P(0 < z < 1.58) \\ &= 0.5 - 0.4429 \ \ (\text{from Appendix B1.}) \\ &= 0.0571 \end{aligned}$$

🎯 Result

a. 12.66

b. 0.7852

c. 0.7281

d. 0.0571

PROBLEM STATEMENT

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Step 1

(a) Selected random numbers: 11, 65, 86, 62, 06, 10, 12, 77, 04

The given ID numbers range from 00 to 24, thus we need to eliminate all random numbers that exceed 24:

$$11, 06, 10, 12, 04$$

The corresponding stores are then in the sample: Formal Man, Summit Stationers, Bootleggers, Leather Ltd., Petries

Step 2

(b) Let us select the 11th row of the random numbers table: Note: Choosing a different row will result in a different sample

82173 26921 28472 98958 07960 66124

We select the first 2-digit number. If the number is between 00 and 24, then we select the corresponding location. If the number is greater than 24, then we ignore the number and select the next 2-digit number.

Repeat until 4 unique locations were selected.

82	$\Rightarrow$	Ignore
17	$\Rightarrow$	Select Dollar Tree
32	$\Rightarrow$	Ignore
69	$\Rightarrow$	Ignore
21	$\Rightarrow$	Select Coach House Gifts
28	$\Rightarrow$	Ignore
47	$\Rightarrow$	Ignore
29	$\Rightarrow$	Ignore
89	$\Rightarrow$	Ignore
58	$\Rightarrow$	Ignore
07	$\Rightarrow$	Select E. B. Brown Opticians
96	$\Rightarrow$	Ignore
06	$\Rightarrow$	Select Summit Stationers

Thus the sample will then contain the four stores: Dollar Tree, Coach House Gifts, E. B. Brown Opticians, Summit Stationers.

Step 3

(c) We need to select every third location with starting point 00 (first point).

Let us first determine the Id numbers that will be in the sample, which are the starting point and the starting point increased by 3 or a multiple of 3.

The given ID numbers range from 00 to 24, thus we need to eliminate all random numbers that exceed 24.

$$
\begin{aligned}
&00\\
00+03=&03\\
03+03=&06\\
06+03=&09\\
09+03=&12\\
12+03=&15\\
15+03=&18\\
18+03=&21\\
21+03=&24\\
24+03=&27
\end{aligned}
$$

The selected random numbers are then 00, 03, 06, 09, 12, 15, 18, 21, 24. The corresponding stores are: Elder-Beerman, Frederick's of Hollywood, Summit Stationers, Lion Store, Leather Ltd., Things Remembered, County Seat, Coach House Gifts, Regis Hairstylists

Result

(a) Formal Man, Summit Stationers, Bootleggers, Leather Ltd., Petries

(b) Answers could vary

(c) Elder-Beerman, Frederick's of Hollywood, Summit Stationers, Lion Store, Leather Ltd., Things Remembered, County Seat, Coach House Gifts, Regis Hairstylists

PROBLEM STATEMENT

Chapter 8, Page 290

Step 1

(a) Selected random numbers: 31, 94, 43, 36, 03, 24, 17, 09

The given ID numbers range from 00 to 38, thus we need to eliminate all random numbers that exceed 38:

$$31, 36, 03, 24, 17, 09$$

The corresponding physicians are then in the sample: Jeanne Fiorito, Douglas Smucker, Jeanine S. Huttner, Harry Mayhew, Mark Steinmetz, Pual Langenkamp

Step 2

(b) Let us select the 11th row of the random numbers table: Note: Choosing a different row will result in a different sample

82173 26921 28472 98958 07960 66124

We select the first 2-digit number. If the number is between 00 and 38, then we select the corresponding location. If the number is greater than 38, then we ignore the number and select the next 2-digit number.

Repeat until 4 unique physicians were selected.

82	$\Rightarrow$	Ignore
17	$\Rightarrow$	Select Mark Steinmetz
32	$\Rightarrow$	Ignore
69	$\Rightarrow$	Ignore
21	$\Rightarrow$	Select J. Christian Zona
28	$\Rightarrow$	Select Ken Bertka
47	$\Rightarrow$	Ignore
29	$\Rightarrow$	Select Mark DeMichiei

Thus the sample will then contain the four physicians: Mark Steinmetz, J. Christian Zona, Ken Bertka, Mark DeMichiei.

Step 3

(c) We need to select every fifth location with starting point 04.

Let us first determine the Id numbers that will be in the sample, which are the starting point and the starting point increased by 5 or a multiple of 5.

The given ID numbers range from 00 to 38, thus we need to eliminate all random numbers that exceed 38.

$$
\begin{aligned}
&04\\
04+05 &= 09\\
09+05 &= 14\\
14+05 &= 19\\
19+05 &= 24\\
24+05 &= 29\\
29+05 &= 34\\
34+05 &= 39
\end{aligned}
$$

The selected random numbers are then 04, 09, 14, 19, 24, 29, and 34. The corresponding physicians are: Francis Aona, Paul Langenkamp, Ricardo Pena, Mark Young, Harry Mayhem, Mark DeMichiei, Richard Koby.

Step 4

(d) We need to select 2 physicians in solo practice (S), 2 in partnership (P), and 1 in group practice (G).

Let us select the 11th row of the random numbers table: Note: Choosing a different row will result in a different sample
82173 26921 28472 98958 07960 66124 89731 95069

We select the first 2-digit number. If the number is between 00 and 38, then we select the corresponding location. If the number is greater than 38, then we ignore the number and select the next 2-digit number.

Repeat until 2 physicians in solo practice (S), 2 in partnership (P), and 1 in group practice (G) were selected (while any additional practices will also be ignored).

82	$\Rightarrow$	Ignore
17	$\Rightarrow$	Select Mark Steinmetz (G)
32	$\Rightarrow$	Ignore
69	$\Rightarrow$	Ignore
21	$\Rightarrow$	Select J. Christian Zona (P)
28	$\Rightarrow$	Ignore (already 1 G selected)
47	$\Rightarrow$	Ignore
29	$\Rightarrow$	Ignore (already 1 G selected)
89	$\Rightarrow$	Ignore
58	$\Rightarrow$	Ignore
07	$\Rightarrow$	Select Judith Furlong (S)
96	$\Rightarrow$	Ignore
06	$\Rightarrow$	Select David DeFrance (S)
61	$\Rightarrow$	Ignore
24	$\Rightarrow$	Select (already 2 S's selected)
89	$\Rightarrow$	Ignore
73	$\Rightarrow$	Ignore
19	$\Rightarrow$	Select Mark Young (P)

Thus the sample will then contain the physicians: Mark Steinmetz, J. Christian Zona, Judith Furlong, David DeFrance, Mark Young.

🎯 Result

(a) Jeanne Fiorito, Douglas Smucker, Jeanine S. Huttner, Harry Mayhew, Mark Steinmetz, Pual Langenkamp

(b) Answers could vary

(c) Francis Aona, Paul Langenkamp, Ricardo Pena, Mark Young, Harry Mayhem, Mark DeMichiei, Richard Koby

(d) Answers could vary

PROBLEM STATEMENT

Chapter 8, Page 291

Step 1

Given population values: 1, 2, 3

(a) We need to list all possible samples of size 2 (including possible repetitions of data values) and compute the mean of each sample.

The sample mean is the sum of all values in the sample divided by the number of data values in the sample.

Sample	**Sample mean**
$1, 1$	$(1+1)/2 = 1$
$1, 2$	$(1+2)/2 = 1.5$
$1, 3$	$(1+3)/2 = 2$
$2, 1$	$(2+1)/2 = 1.5$
$2, 2$	$(2+2)/2 = 2$
$2, 3$	$(2+3)/2 = 2.5$
$3, 1$	$(3+1)/2 = 2$
$3, 2$	$(3+2)/2 = 2.5$
$3, 3$	$(3+3)/2 = 3$

Step 2

(b) Next, we determine the mean of all sample means in part (a):

$$\begin{aligned}&\text{Mean of sample means}\\&= \frac{1+1.5+2+1.5+2+2.5+2+2.5+3}{9}\\&= \frac{18}{9}\\&= 2\end{aligned}$$

The population mean is the sum of all population values divided by the number of population values.

$$\mu = \frac{1+2+3}{3} = \frac{6}{3} = 2$$

We note that the mean of the distribution of the sample mean is equal to the population mean.

Step 3

(c) We note that the data values of the sample mean vary less from the mean of all sample means (due to the values 1.5 and 2.5 being present between 1 and 2, and 2 and 3) than the population values from the population mean.

This then implies that the dispersion of the population is greater than the dispersion of the sample mean.

Step 4

(d) The population distribution is uniform and symmetric, because each of the three population values are equally likely (as they all occur once in the population data set).

The frequency is the number of times the sample mean occurs.

Sample mean	Frequency
1	1
1.5	2
2	3
2.5	2
3	1

We note that the sampling distribution of the sample means is symmetric and approximation bell-shaped, because the highest frequency corresponding to the middle value.

Thus the two distributions are not identical.

Result

(a) Means: 1, 1.5, 2, 1.5, 2, 2.5, 2, 2.5, 3

(b) Both equal to 2

(c) The dispersion of the population is greater than the dispersion of the sample mean.

(d)

Population: Uniform and symmetric

Sample mean: Bell-shaped and uniform

PROBLEM STATEMENT

Chapter 8, Page 291

Step 1

Mean time spent by students on playing organized sports is $\mu = 5.5$ hours per week and the standard deviation is $\sigma = 2.2$ hours per week. Sample size of students is $n = 121$.

Let $\bar{X}$ be the sample mean.

a. Standard error of the mean

$$\begin{aligned}\sigma/\sqrt{n} &= 2.2/\sqrt{121} \\ &= 2.2/11 \\ &= 0.2\end{aligned}$$

$\textbf{b. $\bar{X}$ is between 5 and 6 }$

$$\begin{aligned}P(5 < \bar{X} < 6) &= P(5 < \bar{X} - \mu < 6 - \mu) \\ &= P\left(\frac{5-\mu}{\sigma/\sqrt{n}} < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < \frac{6-\mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(\frac{5-5.5}{2.2/\sqrt{121}} < z < \frac{6-5.5}{2.2/\sqrt{121}}\right) \\ &= P\left(\frac{-0.5}{0.2} < z < \frac{0.5}{0.2}\right) \\ &= P(-2.5 < z < 2.5)\end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-2.5 < z < 0)$, $P(0 < z < 2.5)$ and add the two probabilities.

$$\begin{aligned}P(5 < \bar{X} < 6) &= P(-2.5 < z < 0) + P(0 < z < 2.5) \\ &= P(0 < z < 2.5) + P(0 < z < 2.5) \\ &= 0.4938 + 0.4938 \ \ (\text{from Appendix B1.}) \\ &= 0.9876\end{aligned}$$

Step 2

$\textbf{c. } \bar{X}$ is between 5.3 and 5.7 }$

$$\begin{aligned}
P(5.3 < \bar{X} < 5.7) &= P(5.3 < \bar{X} - \mu < 5.7 - \mu) \\
&= P\left(\frac{5.3 - \mu}{\sigma/\sqrt{n}} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{5.7 - \mu}{\sigma/\sqrt{n}}\right) \\
&= P\left(\frac{5.3 - 5.5}{2.2/\sqrt{121}} < z < \frac{5.7 - 5.5}{2.2/\sqrt{121}}\right) \\
&= P\left(\frac{-0.2}{0.2} < z < \frac{0.2}{0.2}\right) \\
&= P(-1 < z < 1)
\end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-1 < z < 0)$, $P(0 < z < 1)$ and add the two probabilities.

$$\begin{aligned}
P(5.3 < \bar{X} < 5.7) &= P(-1 < z < 0) + P(0 < z < 1) \\
&= P(0 < z < 1) + P(0 < z < 1) \\
&= 0.3413 + 0.3413 \ \ (\text{from Appendix B1.}) \\
&= 0.6826
\end{aligned}$$

Step 3

$\textbf{d. } \bar{X}$ is greater than 6.5 }$

$$\begin{aligned}
P(\bar{X} > 6.5) &= P(\bar{X} - \mu > 6.5 - \mu) \\
&= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{6.5 - \mu}{\sigma/\sqrt{n}}\right) \\
&= P\left(z > \frac{6.5 - 5.5}{2.2/\sqrt{121}}\right) \\
&= P\left(z > \frac{1}{0.2}\right) \\
&= P(z > 5) \\
&= 0.5 - P(0 < z < 5) \\
&= 0.5 - 0.5 \\
&\ (\text{Probability of } z \text{ lying between } 0 \text{ and } z_1 \text{ is } 0.5 \text{ when } z_1 \text{ is above } 3) \\
&= 0
\end{aligned}$$

Result

a. 0.2

b. 0.9876

c. 0.6826

d. 0

Chapter 8

Exercise 23

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PROBLEM STATEMENT

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Step 1

Given:

Martketing-Gets-Results: 50 randomly selected consumers who plan to purchase laptop within the next year

Marketing-Reaps-Profits: 200 current laptop owners

The larger the sample, the more information about the population that is present in the sample and thus the more accurate the results of the sample will be.

This then implies that the sample of 200 currently laptop owners will be more useful as its results are more accurate.

Moreover, this sample contains people that are already familiar with laptops, which is not necessarily true for the people in the sample of 50 consumers.

Result

Marketing-Reaps-Profits: 200 current laptop owners

PROBLEM STATEMENT

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Step 1

(a) The standard error of the mean measures the variability in the sample means.

When the sample size is increased, then we have more information about the population and thus the estimates from the sample will be more accurate.

However, more accurate results implies that the variability decreases and thus the standard error of the mean **decreases**.

Step 2

(b) By the central limit theorem, the distribution of the sample means is approximately normal when the sample is large.

Moreover, the distribution of the sample means becomes closer and closer to the normal distribution as the sample size increases.

Step 3

(c) When the sample size is increased, then we have more information about the population and thus the estimates from the sample will be more accurate.

This then implies that the benefit of using large sample sizes is that the estimation of the population mean is closer to the true value of the population mean.

Result

(a) Decreases

(b) Shape becomes closer to the shape of the normal distribution

(c) The estimation of the population mean is closer to the true value of the population mean.

PROBLEM STATEMENT

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Step 1

(a) Let us select the 11th row of the random numbers table: Note: Choosing a different row will result in a different sample

82173 26921 28472 98958 07960 66124

We select the first 2-digit number. If the number is between 01 and 25, then we select the corresponding number (that is, we select the xth number when the random number is x). If the number is greater than 25, then we ignore the number and select the next 2-digit number.

Repeat until 5 unique motels were selected.

82	$\Rightarrow$	Ignore
17	$\Rightarrow$	Select 104
32	$\Rightarrow$	Ignore
69	$\Rightarrow$	Ignore
21	$\Rightarrow$	Select 60
28	$\Rightarrow$	Ignore
47	$\Rightarrow$	Ignore
29	$\Rightarrow$	Ignore
89	$\Rightarrow$	Ignore
58	$\Rightarrow$	Ignore
07	$\Rightarrow$	Select 84
96	$\Rightarrow$	Ignore
06	$\Rightarrow$	Select 72
61	$\Rightarrow$	Ignore
24	$\Rightarrow$	Select 58

Thus the sample will then contain the data values: 104, 60, 84, 72, 58.

Step 2

(b) We need to select every fifth data value with starting point x.

x needs to be randomly selected from the first five motels, I will choose the second motel (as 2 is the first digit from 1 to 5 mentioned in row 11). Note: Choosing a different row will result in a different sample

Let us first determine the random numbers that will represent the data values in the sample, which are the starting point and the starting point increased by 5 or a multiple of 5.

We need to eliminate all random numbers that exceed 25, because we have only been given 25 data values.

$$
\begin{aligned}
&02\\
02+05 =&07\\
07+05 =&12\\
12+05 =&17\\
17+05 =&22\\
22+05 =&27
\end{aligned}
$$

The selected random numbers are then 02, 07, 12, 17, and 22. The corresponding data values are: 72, 84, 115, 104, and 80.

Step 3

(c) We need to select 3 regular motels and two cut-rate motels (where the last five motels are cut-rate motels).

Let us select the 11th row of the random numbers table: Note: Choosing a different row will result in a different sample

82173 26921 28472 98958 07960 66124

We select the first 2-digit number. If the number is between 01 and 25, then we select the corresponding number (that is, we select the xth number when the random number is x). If the number is greater than 25, then we ignore the number and select the next 2-digit number.

Repeat until 3 regular motels (R) and 2 cut-rate motels (C) were selected.

82	$\Rightarrow$	Ignore
17	$\Rightarrow$	Select 104 (R)
32	$\Rightarrow$	Ignore
69	$\Rightarrow$	Ignore
21	$\Rightarrow$	Select 60 (C)
28	$\Rightarrow$	Ignore
47	$\Rightarrow$	Ignore
29	$\Rightarrow$	Ignore
89	$\Rightarrow$	Ignore
58	$\Rightarrow$	Ignore
07	$\Rightarrow$	Select 84 (R)
96	$\Rightarrow$	Ignore
06	$\Rightarrow$	Select 72 (R)
61	$\Rightarrow$	Ignore
24	$\Rightarrow$	Select 58 (C)

Thus the sample will then contain the data values: 104, 60, 84, 72, 58.

Result

Answers could vary

PROBLEM STATEMENT

Chapter 8, Page 292

Step 1

(a) Let us select the 11th row of the random numbers table: $\text{\color{#4257b2}Note: Choosing a different row will result in a different sample\color{default}\

82173 26921 28472 98958 07960 66124 89731 95069 18625 92405

97594 25168 89178 68190 05043 17407 48201 83917 11413 7220\

We select the first 3-digit number. If the number is between 001 and 250, then we select the corresponding number. We ignore all other numbers.\

Repeat until 10 unique numbers were selected.}$

Step 2

$$
\begin{aligned}
821 &\Rightarrow \text{Ignore} \\
732 &\Rightarrow \text{Ignore} \\
692 &\Rightarrow \text{Ignore} \\
128 &\Rightarrow \text{Select } 128 \\
472 &\Rightarrow \text{Ignore} \\
989 &\Rightarrow \text{Ignore} \\
580 &\Rightarrow \text{Ignore} \\
796 &\Rightarrow \text{Ignore} \\
066 &\Rightarrow \text{Select } 066 \\
124 &\Rightarrow \text{Select } 124 \\
897 &\Rightarrow \text{Ignore} \\
319 &\Rightarrow \text{Ignore} \\
506 &\Rightarrow \text{Ignore} \\
918 &\Rightarrow \text{Ignore} \\
625 &\Rightarrow \text{Ignore} \\
924 &\Rightarrow \text{Ignore} \\
059 &\Rightarrow \text{Select } 059 \\
759 &\Rightarrow \text{Ignore} \\
425 &\Rightarrow \text{Ignore} \\
168 &\Rightarrow \text{Select } 168 \\
891 &\Rightarrow \text{Ignore} \\
786 &\Rightarrow \text{Ignore} \\
819 &\Rightarrow \text{Ignore} \\
005 &\Rightarrow \text{Select } 005 \\
043 &\Rightarrow \text{Select } 043 \\
174 &\Rightarrow \text{Select } 174 \\
074 &\Rightarrow \text{Select } 074 \\
820 &\Rightarrow \text{Ignore} \\
183 &\Rightarrow \text{Select } 183
\end{aligned}
$$

Thus the 10 usable numbers are then: 128, 066, 124, 059, 168, 005, 043, 174, 074, 183.

Step 3

(b) We need to select every 25th number with a starting point between 001 and 025 (as then we will select 10 numbers from 001 to 250).

Let us choose the starting point as the first number in part (a) that was from 001 to 025, which ended up being 005. Note: A different starting point leads to a different sample.

Let us first determine the numbers that will be in the sample, which are the starting point and the starting point increased by 25 or a multiple of 25.

The given numbers range from 001 to 250, thus we need to eliminate all random numbers that exceed 250.

$$
\begin{aligned}
&005\\
005+25 &= 030\\
030+25 &= 055\\
055+25 &= 080\\
080+25 &= 105\\
105+25 &= 130\\
130+25 &= 155\\
155+25 &= 180\\
180+25 &= 205\\
205+25 &= 230\\
230+25 &= 255
\end{aligned}
$$

The selected random numbers are then 005, 030, 055, 080, 105, 130, 155, 180, 205, 230, 255.

Step 4

(c) The advantage of the method in part (a) is that the data values are more randomly selected from the population than the method in part (b), which will lead to more accurate results.

The advantage of the method in part (b) is that it is much easier and much less time-consuming to determine the data values in the sample (in comparison with the method in part (a)).

Step 5

(d) There are many different possible ways in which to select 10 passengers randomly.

For example, when the passengers are either in first class or in second class, then we could choose to select 8 passengers randomly in first class and to select 2 passengers randomly in third class.

🎯 Result

(a)-(b) Answers could vary

(c) Part (a) more random, part (b) easier and less-time consuming

(d) Answers could vary

PROBLEM STATEMENT

Chapter 8, Page 292

Step 1

Given data values: 79, 64, 84, 82, 92, 77

(a) We need to select 2 of the 6 grades.

$$n = 6$$
$$r = 2$$

Since the order of the grades does not matter (as a different order of the selected grades results in the same sample), we need to use the definition of a combination:

$$ {}_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!} $$

$$ {}_6C_2 = \frac{6!}{2!(6-2)!} = \frac{6!}{2!4!} = 15 $$

Thus there are 15 possible samples of size 2.

Step 2

(b) We need to determine all possible samples of size 2 and determine its sample mean.

The sample mean is the sum of all data values divided by the number of data values.

Sample	Sample mean
$79, 64$	$(79+64)/2 = 71.5$
$79, 84$	$(79+84)/2 = 81.5$
$79, 82$	$(79+82)/2 = 80.5$
$79, 92$	$(79+92)/2 = 85.5$
$79, 77$	$(79+77)/2 = 78$
$64, 84$	$(64+84)/2 = 74$
$64, 82$	$(64+82)/2 = 73$
$64, 92$	$(64+92)/2 = 78$
$64, 77$	$(64+77)/2 = 70.5$
$84, 82$	$(84+82)/2 = 83$
$84, 92$	$(84+92)/2 = 88$
$84, 77$	$(84+77)/2 = 80.5$
$82, 92$	$(82+92)/2 = 87$
$82, 77$	$(82+77)/2 = 79.5$
$92, 77$	$(92+77)/2 = 84.5$

Step 3

(c) We need to determine the mean of all sample means in part (b):

$$\begin{aligned}\mu_{\overline{X}} &= \frac{\begin{array}{c}71.5+81.5+80.5+85.5+78\\+74+73+78+70.5+83\\+88+80.5+87+79.5+84.5\end{array}}{15}\\ &= \frac{1195}{15}\\ &\approx 79.66667\end{aligned}$$

The population mean is the sum of all population values divided by the number of values:

$$\mu = \frac{\sum x_i}{N} = \frac{79+64+84+82+92+77}{6} = \frac{478}{6} \approx 79.6667$$

We note that the population mean and the mean of the sampling distribution of the sample mean are identical.

Step 4

(d) I would not like this arrangement, because I would be guaranteed a final percent of 79.6667% when including all six data values and I have 7 chances out of 15 to score less with this arrangement (which is very likely to occur). Moreover, it's possible that the percentage will drop as low as 70.5.

The result would be different when we drop the lowest score, because then we will most likely have a much higher percentage.

🎯 Result

(a) 15 different samples

(b) Sample means: 71.5, 81.5, 80.5, 85.5, 78, 74, 73, 78, 70.5, 83, 88, 80.5, 87, 79.5, 84.5

(c) Both equal to 79.6667

(d) No, Yes

PROBLEM STATEMENT

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Step 1

Number of errors made by tellers are : $2, 3, 5, 3$ and 5.

a.

Total number of tellers in the population is $N = 5$ and the sample size is $n = 2$ Total number of possible samples of size 2 is

$$\begin{aligned} {}_N C_n &= \frac{N!}{n!(N-n)!} \\ &= \frac{5!}{2!(5-2)!} \\ &= \frac{5!}{2!3!} \\ &= 10 \end{aligned}$$

Let X_i be the ith sample of size 2. So,

$$\bar{X}_i = \frac{X_{i1} + X_{i2}}{2}$$

b. Sample means for all possible samples of size 2 is given in the below table.

Step 2

Sample	Values	Sum $\sum_j X_{ij}$	Mean $\bar{X}_i$
1	$2, 3$	5	2.5
2	$2, 5$	7	3.5
3	$2, 3$	5	2.5
4	$2, 5$	7	3.5
5	$3, 5$	8	4
6	$3, 3$	6	3
7	$3, 5$	8	4
8	$5, 3$	8	4
9	$5, 5$	10	5
10	$3, 5$	8	4

c. Population mean is μ.

$$\begin{aligned}\mu &= \frac{2+3+5+3+5}{5} \\ &= \frac{18}{5} \\ &= 3.6\end{aligned}$$

Sampling distribution of the sample mean for $n = 2$ is as follows.

Sample Mean	Number of means	Probability
2.5	2	0.2
3	1	0.1
3.5	2	0.2
4	4	0.4
5	1	0.1

The sample mean is $\mu_{\bar{X}}$

$$\begin{aligned}\mu_{\bar{X}} &= \frac{\text{sum of all sample means}}{\text{total number of samples}} \\ &= \frac{2.5 \times 2 + 3 \times 1 + 3.5 \times 2 + 4 \times 4 + 5 \times 1}{10} \\ &= \frac{5+3+7+16+5}{10} \\ &= \frac{36}{10} \\ &= 3.6\end{aligned}$$

This shows that $\mu = \mu_{\bar{X}}$.

Result

a. 10 samples

c. $\mu = 3.6$ and $\mu_{\bar{X}} = 3.6$

PROBLEM STATEMENT

Chapter 8, Page 292

Step 1

(a) We need to select 2 of the 5 technicians.

$$n = 5$$
$$r = 2$$

The order of the technicians does not matter (as a different order of the selected technicians results in the same sample), thus we need to use the definition of a combination:

$$_5C_2 = \frac{5!}{2!(5-2)!} = \frac{5!}{2!3!} = 10$$

Thus there are 10 different samples.

Step 2

(b) We need to list all possible samples of two observations and compute the sample mean of each of the samples.

The sample mean is the sum of all data values in the sample divided by the number of data values.

Sample	Sample mean
$Taylor, Hurley$	$(4+3)/2 = 3.5$
$Taylor, Gupta$	$(4+5)/2 = 4.5$
$Taylor, Rousche$	$(4+3)/2 = 3.5$
$Taylor, Huang$	$(4+2)/2 = 3$
$Hurley, Gupta$	$(3+5)/2 = 4$
$Hurley, Rousche$	$(3+3)/2 = 3$
$Hurley, Huang$	$(3+2)/2 = 2.5$
$Gupta, Rousche$	$(5+3)/2 = 4$
$Gupta, Huang$	$(5+2)/2 = 3.5$
$Rousche, Huang$	$(3+2)/2 = 2.5$

Step 3

(c) Next, we need to determine the mean of all sample means in part (b):

$$\begin{aligned}\mu_{\overline{X}} &= \frac{\begin{matrix}3.5+4.5+3.5+3+4\\+3+2.5+4+3.5+2.5\end{matrix}}{10}\\ &= \frac{34}{10}\\ &= 3.4\end{aligned}$$

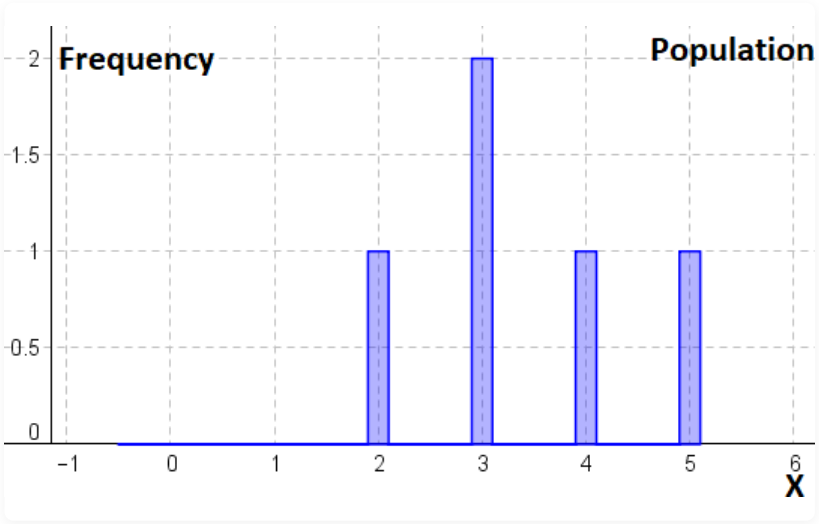
The population mean is the sum of all population value divided by the number of population values.

$$\mu = \frac{\sum x_i}{N} = \frac{4+3+5+3+2}{5} = \frac{17}{5} = 3.4$$

We note that the population mean is equal to the mean of all sample means.

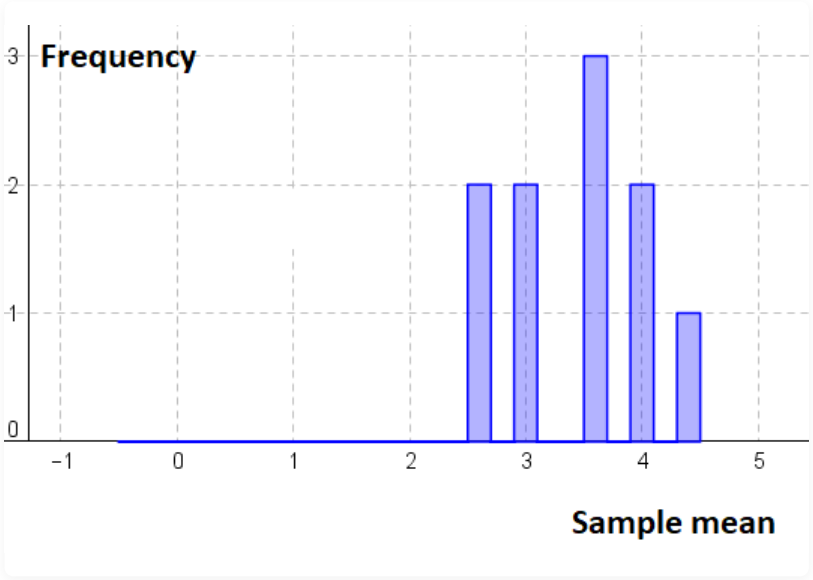
Step 4

(d) We create a histogram for both the population and for all sample means in part (b), where the population values or sample mean are on the horizontal axis, the bars are equally wide, and the height of the bars is equal to the number of times the particular data value occurs.



We note that both distributions have approximately the same shape with a single peak, although the distribution of the sample means appears to be a bit more symmetric than the population distribution.

Step 5



Result

(a) 10 different samples

(b) Sample means: 3.5, 4.5, 3.5, 3, 4, 3, 2.5, 4, 3.5, 2.5

(c) Both equal to 3.4

PROBLEM STATEMENT

Chapter 8, Page 292

Step 1

(a) We need to select 2 of the 6 sales representative.

$$n = 6$$
$$r = 2$$

The order of the sales representatives does not matter (as a different order of the selected sales representatives results in the same sample), thus we need to use the definition of a combination:

$$_6C_2 = \frac{6!}{2!(6-2)!} = \frac{6!}{2!4!} = 15$$

Thus there are 15 different samples.

Step 2

(b) We need to list all possible samples of two observations and compute the sample mean of each of the samples.

The sample mean is the sum of all data values in the sample divided by the number of data values.

Sample	Sample mean
$Zina, Woon$	$(54+50)/2=52$
$Zina, Ernie$	$(54+52)/2=53$
$Zina, Jan$	$(54+48)/2=51$
$Zina, Molly$	$(54+50)/2=52$
$Zina, Rachel$	$(54+52)/2=53$
$Woon, Ernie$	$(50+52)/2=51$
$Woon, Jan$	$(50+48)/2=49$
$Woon, Molly$	$(50+50)/2=50$
$Woon, Rachel$	$(50+52)/2=51$
$Ernie, Jan$	$(52+48)/2=50$
$Ernie, Molly$	$(52+50)/2=51$
$Ernie, Rachel$	$(52+52)/2=52$
$Jan, Molly$	$(48+50)/2=49$
$Jan, Rachel$	$(48+52)/2=50$
$Molly, Rachel$	$(50+52)/2=51$

Step 3

(c) The frequency is the number of times the sample mean occurs in part (b):

Sample mean	Frequency
49	2
50	3
51	5
52	3
53	2

Step 4

(d) Next, we need to determine the mean of all sample means in part (b):

$$\begin{aligned}\mu_{\overline{X}} &= \frac{\begin{array}{c}52+53+51+52+53\\+51+49+50+51+50\\+51+52+49+50+51\end{array}}{15}\\ &= \frac{765}{15}\\ &= 51\end{aligned}$$

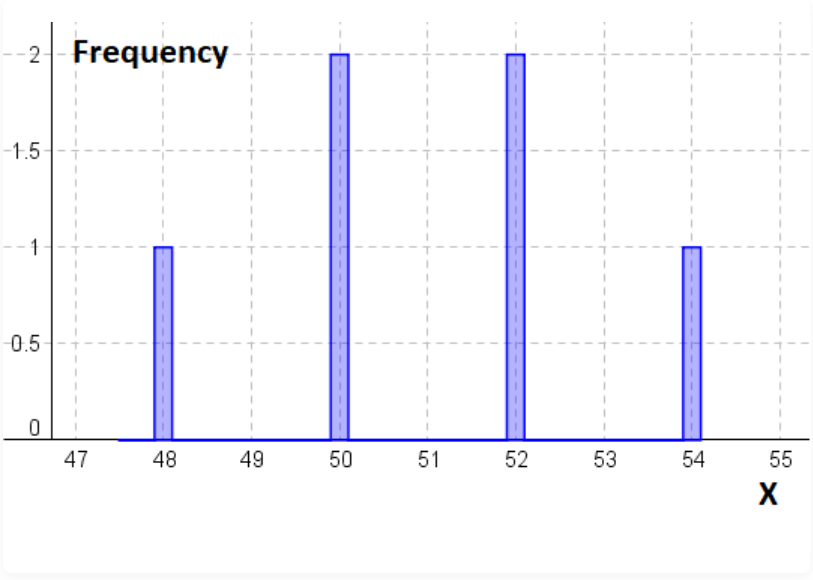
The population mean is the sum of all population value divided by the number of population values.

$$\mu = \frac{\sum x_i}{N} = \frac{4+50+52+48+50+52}{6} = \frac{306}{6} = 51$$

We note that the population mean is equal to the mean of all sample means.

Step 5

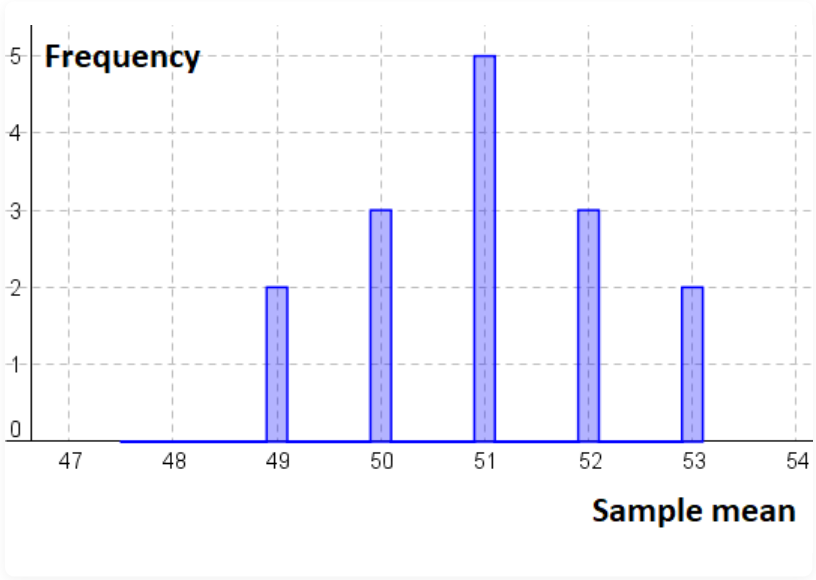
(e) We create a histogram for the population distribution, where the population values are on the horizontal axis, the bars are equally wide, and the height of the bars is equal to the number of times the particular population value occurs.



We note that the distribution is symmetric and approximately bell-shaped, because the highest bars are present in the middle of the distribution.

Step 6

(f) We create a histogram for all sample means in part (b), where the sample means are on the horizontal axis, the bars are equally wide, and the height of the bars is equal to the number of times the particular sample mean occurs.



We note that the distribution is symmetric and approximately bell-shaped, because the highest bars are present in the middle of the distribution.

Result

(a) 15 different samples

(b) Sample means: 52, 53, 51, 52, 53, 51, 49, 50, 51, 50, 51, 52, 49, 50, 51

(c)

Sample mean	Frequency
49	2
50	3
51	5
52	3
53	2

(d) 51, 51

(e) Symmetric and approximately bell-shaped

(f) Symmetric and approximately bell-shaped

PROBLEM STATEMENT

Chapter 8, Page 293

Step 1

Mean life of AA batteries is $\mu = 35$ hours and the standard deviation is $\sigma = 5.5$ hours. Sample size is $n = 25$.

Let $\bar{X}$ be the sample mean.

a. By the Central Limit Theorem, the distribution of the sample mean will be a normal distribution.

b. Standard error of the mean

$$\begin{aligned}\sigma/\sqrt{n} &= 5.5/\sqrt{25} \\ &= 5.5/5 \\ &= 1.1\end{aligned}$$

Step 2

$\textbf{c. } \bar{X}$ is greater than 36 }$

$$\begin{aligned}P(\bar{X} > 36) &= P(\bar{X} - \mu > 36 - \mu) \\ &= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{36 - \mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(z > \frac{36 - 35}{5.5/\sqrt{25}}\right) \\ &= P\left(z > \frac{1}{1.1}\right) \\ &= P(z > 0.91) \\ &= 0.5 - P(0 < z < 0.91) \\ &= 0.5 - 0.3186 \ \ (\text{from Appendix B1.}) \\ &= 0.1814\end{aligned}$$

18.14% of the samples ill have a useful life of more than 36 hours.

Step 3

$\textbf{d. } \bar{X}$ is greater than 34.5 }$

$$\begin{aligned}
P(\bar{X} > 34.5) &= P(\bar{X} - \mu > 34.5 - \mu) \\
&= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{34.5 - \mu}{\sigma/\sqrt{n}}\right) \\
&= P\left(z > \frac{34.5 - 35}{5.5/\sqrt{25}}\right) \\
&= P\left(z > \frac{-0.5}{1.1}\right) \\
&= P(z > -0.45) \\
&= P(z < 0.45) \\
&= 0.5 + P(0 < z < 0.45) \\
&= 0.5 - 0.1736 \ \text{(from Appendix B1.)} \\
&= 0.3264
\end{aligned}$$

32.64% of the samples ill have a useful life of more than 34.5 hours.

Step 4

$\textbf{e. } \bar{X}$ is between 34.5 and 36 }$

$$\begin{aligned}
P(34.5 < \bar{X} < 36) &= P(34.5 < \bar{X} - \mu < 36 - \mu) \\
&= P\left(\frac{34.5 - \mu}{\sigma/\sqrt{n}} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{36 - \mu}{\sigma/\sqrt{n}}\right) \\
&= P\left(\frac{34.5 - 35}{5.5/\sqrt{25}} < z < \frac{36 - 35}{5.5/\sqrt{25}}\right) \\
&= P\left(\frac{-0.5}{1.1} < z < \frac{1}{1.1}\right) \\
&= P(-0.45 < z < 0.91)
\end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-0.45 < z < 0)$, $P(0 < z < 0.91)$ and add the two probabilities.

$$\begin{aligned}
P(34.5 < \bar{X} < 36) &= P(-0.45 < z < 0) + P(0 < z < 0.91) \\
&= P(0 < z < 0.45) + P(0 < z < 0.91) \\
&= 0.1736 + 0.3186 \ \text{(from Appendix B1.)} \\
&= 0.4922
\end{aligned}$$

49.22% of the samples ill have a useful life between 34.5 and 36 hours.

🎯 Result

a. Normal Distribution

b. 1.1

c. 18.14%

d. 32.64%

e. 49.22%

PROBLEM STATEMENT

Chapter 8, Page 293

Step 1

Mean lengths of the cuts on a CD is $\mu = 135$ seconds and the standard deviation is $\sigma = 8$ seconds. Sample size or cuts is $n = 16$.

Let $\bar{X}$ be the sample mean.

a. By the Central Limit Theorem, the distribution of the sample mean will be a normal distribution.

b. Standard error of the mean

$$\begin{aligned}\sigma/\sqrt{n} &= 8/\sqrt{16} \\ &= 8/4 \\ &= 2\end{aligned}$$

Step 2

$\textbf{c. $\bar{X}$ is greater than 140 }$

$$\begin{aligned}P(\bar{X} > 140) &= P(\bar{X} - \mu > 140 - \mu) \\ &= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{140 - \mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(z > \frac{140 - 135}{8/\sqrt{16}}\right) \\ &= P\left(z > \frac{5}{2}\right) \\ &= P(z > 2.5) \\ &= 0.5 - P(0 < z < 2.5) \\ &= 0.5 - 0.4938 \ \ (\text{from Appendix B1.}) \\ &= 0.0062\end{aligned}$$

0.62% of the samples will be greater than 140 seconds.

Step 3

$\textbf{d. } \bar{X}$ is greater than 128 }$

$$
\begin{aligned}
P(\bar{X} > 128) &= P(\bar{X} - \mu > 128 - \mu) \\
&= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{128 - \mu}{\sigma/\sqrt{n}}\right) \\
&= P\left(z > \frac{128 - 135}{8/\sqrt{16}}\right) \\
&= P\left(z > \frac{-7}{2}\right) \\
&= P(z > -3.5) \\
&= P(z < 3.5) \\
&= 0.5 + P(0 < z < 3.5) \\
&= 0.5 + 0.5 \\
&\text{(Probability of } z \text{ between 0 and } z_1 \text{ is 0.5 if } z_1 \text{ is greater than 3.1)} \\
&= 1
\end{aligned}
$$

All samples will be greater than 128 seconds.

Step 4

$\textbf{e. } \bar{X}$ is between 128 and 140 }$

$$
\begin{aligned}
P(128 < \bar{X} < 140) &= P(128 < \bar{X} - \mu < 140 - \mu) \\
&= P\left(\frac{128 - \mu}{\sigma/\sqrt{n}} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{140 - \mu}{\sigma/\sqrt{n}}\right) \\
&= P\left(\frac{128 - 135}{8/\sqrt{16}} < z < \frac{140 - 135}{8/\sqrt{16}}\right) \\
&= P\left(\frac{-7}{2} < z < \frac{5}{2}\right) \\
&= P(-3.5 < z < 2.5)
\end{aligned}
$$

The two points are on the different sides of the mean, so calculate $P(-3.5 < z < 0)$, $P(0 < z < 2.5)$ and add the two probabilities.

$$
\begin{aligned}
P(128 < \bar{X} < 140) &= P(-3.5 < z < 0) + P(0 < z < 2.5) \\
&= P(0 < z < 3.5) + P(0 < z < 2.5) \\
&= 0.5 + 0.4938 \ \text{(from Appendix B1.)} \\
&= 0.9938
\end{aligned}
$$

99.38% of the samples will be greater than 128 seconds but less than 140 seconds.

🎯 Result

a. Normal Distribution

b. 2

c. 0.62%

d. 100%

e. 99.38%

PROBLEM STATEMENT

Chapter 8, Page 293

Step 1

Mean amount spent by women on personal products is $\mu = \$350$ per year and the standard deviation is $\sigma = \$45$ per year. Sample size is $n = 40$. Let $\bar{X}$ be the sample mean.

$\bar{X}$ is greater than 335

$$\begin{aligned}
P(\bar{X} > 335) &= P(\bar{X} - \mu > 335 - \mu) \\
&= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{335 - \mu}{\sigma/\sqrt{n}}\right) \\
&= P\left(z > \frac{335 - 350}{45/\sqrt{40}}\right) \\
&= P\left(z > \frac{-15}{7.12}\right) \\
&= P(z > -2.11) \\
&= P(z < 2.11) \\
&= 0.5 + P(0 < z < 2.11) \\
&= 0.5 + 0.4826 \quad \text{(from Appendix B1.)} \\
&= 0.9826
\end{aligned}$$

Result

$$0.9826$$

PROBLEM STATEMENT

Chapter 8, Page 293

Step 1

Mean amount of life insurance per household is $\mu = \$110,000$ and the standard deviation is $\sigma = \$40,000$.
Sample of households is $n = 50$.

Let $\bar{X}$ be the sample mean.

a. Standard error of the mean

$$\begin{aligned}\sigma/\sqrt{n} &= 40000/\sqrt{50} \\ &= 40000/7.07 \\ &= 5658\end{aligned}$$

b. By the Central Limit Theorem, the distribution of the sample mean will be a normal distribution.

Step 2

$\textbf{c. } \bar{X}$ is greater than 112000 }$

$$\begin{aligned}P(\bar{X} > 112000) &= P(\bar{X} - \mu > 112000 - \mu) \\ &= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{112000 - \mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(z > \frac{112000 - 110000}{40000/\sqrt{50}}\right) \\ &= P\left(z > \frac{2000}{5658}\right) \\ &= P(z > 0.35) \\ &= 0.5 - P(0 < z < 0.35) \\ &= 0.5 - 0.1368 \ \ (\text{from Appendix B1.}) \\ &= 0.3632\end{aligned}$$

Step 3

$\textbf{d. } \bar{X} \text{ is greater than 100000 }$

$$\begin{aligned}
P(\bar{X} > 100000) &= P(\bar{X} - \mu > 100000 - \mu) \\
&= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{100000 - \mu}{\sigma/\sqrt{n}}\right) \\
&= P\left(z > \frac{100000 - 110000}{40000/\sqrt{50}}\right) \\
&= P\left(z > \frac{-10000}{5658}\right) \\
&= P(z > -1.78) \\
&= P(z < 1.78) \\
&= 0.5 + P(0 < z < 1.78) \\
&= 0.5 + 0.4625 \ \text{(from Appendix B1.)} \\
&= 0.9625
\end{aligned}$$

Step 4

$\textbf{e. } \bar{X} \text{ is between 100000 and 112000 }$

$$\begin{aligned}
P(100000 < \bar{X} < 112000) &= P(100000 < \bar{X} - \mu < 112000 - \mu) \\
&= P\left(\frac{100000 - \mu}{\sigma/\sqrt{n}} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{112000 - \mu}{\sigma/\sqrt{n}}\right) \\
&= P\left(\frac{100000 - 110000}{40000/\sqrt{50}} < z < \frac{112000 - 110000}{40000/\sqrt{50}}\right) \\
&= P\left(\frac{-10000}{5658} < z < \frac{2000}{5658}\right) \\
&= P(-1.78 < z < 0.35)
\end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-1.78 < z < 0)$, $P(0 < z < 0.35)$ and add the two probabilities.

$$\begin{aligned}
P(100000 < \bar{X} < 112000) &= P(-1.78 < z < 0) + P(0 < z < 0.35) \\
&= P(0 < z < 1.78) + P(0 < z < 0.35) \\
&= 0.4625 + 0.1368 \ \text{(from Appendix B1.)} \\
&= 0.5993
\end{aligned}$$

🎯 **Result**

a. 5658

b. Normal Distribution

c. 0.3632

d. 0.9625

e. 0.5993

PROBLEM STATEMENT

Chapter 8, Page 293

Step 1

The mean age at which men in the United States marry for the first time is $\mu = 24.8$ years and the standard deviation is $\sigma = 2.5$ years. Sample size of men is $n = 60$.

Let $\bar{X}$ be the sample mean.

$\textbf{b. } \bar{X}$ is less than 25.1 }$

$$\begin{aligned} P(\bar{X} < 25.1) &= P(\bar{X} - \mu < 25.1 - \mu) \\ &= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{25.1 - \mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(z < \frac{25.1 - 24.8}{2.5/\sqrt{60}}\right) \\ &= P\left(z < \frac{0.3}{0.322}\right) \\ &= P(z < 0.93) \\ &= 0.5 + P(0 < z < 0.93) \\ &= 0.5 + 0.3238 \ \ (\text{from Appendix B1.}) \\ &= 0.8238 \end{aligned}$$

Result

$$0.8238$$

PROBLEM STATEMENT

Chapter 8, Page 293

Step 1

The mean fare charged by taxi drivers is $\mu = \$21$ and the standard deviation is $\sigma = \$3.50$. Sample size of fares is $n = 15$.

Let $\bar{X}$ be the sample mean.

$\textbf{a. } \bar{X}$ is between 20 and 23 }$

$$\begin{aligned} P(20 < \bar{X} < 23) &= P(20 < \bar{X} - \mu < 23 - \mu) \\ &= P\left(\frac{20-\mu}{\sigma/\sqrt{n}} < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < \frac{23-\mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(\frac{20-21}{3.5/\sqrt{15}} < z < \frac{23-21}{3.5/\sqrt{15}}\right) \\ &= P\left(\frac{-1}{0.904} < z < \frac{2}{0.904}\right) \\ &= P(-1.11 < z < 2.21) \end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-1.11 < z < 0)$, $P(0 < z < 2.21)$ and add the two probabilities.

$$\begin{aligned} P(20 < \bar{X} < 23) &= P(-1.11 < z < 0) + P(0 < z < 2.21) \\ &= P(0 < z < 1.11) + P(0 < z < 2.21) \\ &= 0.3665 + 0.4864 \ \ (\text{from Appendix B1.}) \\ &= 0.8529 \end{aligned}$$

b. One needs to assume that the distribution of fares follow normal distribution. Assuming fares are normally distributed, the sample mean also follows normal distribution using Central Limit Theorem.

Result

a. 0.8529

b. Assume Normal Distribution

PROBLEM STATEMENT

Chapter 8, Page 293

Step 1

The mean weight of delivery trucks is $\mu = 6000$ pounds and the standard deviation is $\sigma = 150$ pounds. Sample size of trucks is $n = 40$.

Let $\bar{X}$ be the sample mean.

About 95% of the area under the normal curve is within 1.96 standard deviation of the mean i.e $\mu \pm 1.96\sigma/\sqrt{n}$

$$\mu + 1.96\sigma/\sqrt{n} = 6000 + 1.96(150/\sqrt{40}) = 6000 + 1.96 \times 23.73 = 6046.51$$

and

$$\mu - 1.96\sigma/\sqrt{n} = 6000 - 1.96(150/\sqrt{40}) = 6000 - 1.96 \times 23.73 = 5953.5$$

About 95% of the sample mean lie between 5953.5 pounds and 6046.5 pounds.

Result

5953.5 pounds and 6046.5 pounds

PROBLEM STATEMENT

Chapter 8, Page 293

Step 1

The mean amount purchased by a customer is $\mu = \$23.5$ and the standard deviation is $\sigma = \$5$. Sample size of customers is $n = 50$.

Let $\bar{X}$ be the sample mean.

$\textbf{a. $\bar{X}$ is greater than 25 }$

$$\begin{aligned} P(\bar{X} > 63) &= P(\bar{X} - \mu > 25 - \mu) \\ &= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{25 - \mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(z > \frac{25 - 23.5}{5/\sqrt{50}}\right) \\ &= P\left(z > \frac{1.5}{0.707}\right) \\ &= P(z > 2.12) \\ &= 0.5 - P(0 < z < 2.12) \\ &= 0.5 - 0.4830 \ \ (\text{from Appendix B1.}) \\ &= 0.017 \end{aligned}$$

Step 2

$\textbf{b. } \bar{X}$ is between 22.5 and 25 }$

$$\begin{aligned} P(22.5 < \bar{X} < 25) &= P(22.5 < \bar{X} - \mu < 25 - \mu) \\ &= P\left(\frac{22.5 - \mu}{\sigma/\sqrt{n}} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{25 - \mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(\frac{22.5 - 23.5}{5/\sqrt{50}} < z < \frac{25 - 23.5}{5/\sqrt{50}}\right) \\ &= P\left(\frac{-1}{0.707} < z < \frac{1.5}{0.707}\right) \\ &= P(-1.41 < z < 2.12) \end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-1.41 < z < 0)$, $P(0 < z < 2.12)$ and add the two probabilities.

$$\begin{aligned} P(22.5 < \bar{X} < 25) &= P(-1.41 < z < 0) + P(0 < z < 2.12) \\ &= P(0 < z < 1.41) + P(0 < z < 2.12) \\ &= 0.4207 + 0.4830 \ \ (\text{from Appendix B1.}) \\ &= 0.9037 \end{aligned}$$

Step 3

c. About 90% of the area under the normal curve is within 1.64 standard deviation of the mean i.e $\mu \pm 1.64\sigma/\sqrt{n}$

$$\mu + 1.64\sigma/\sqrt{n} = 23.5 + 1.64(5/\sqrt{50}) = 23.5 + 1.64 \times 0.707 = 24.66$$

and

$$\mu - 1.64\sigma/\sqrt{n} = 23.5 - 1.64(5/\sqrt{50}) = 23.5 - 1.64 \times 0.707 = 22.34$$

About 90% of the observations lie between $\$22.34$ and $\$24.66$.

Result

a. 0.017

b. 0.9037

c. $\$22.34$ and $\$24.66$

Chapter 8 **Exercise 39** Page 293

PROBLEM STATEMENT

Chapter 8, Page 293

Step 1

The mean SAT score for Division I student-athletes is $\mu = 947$ and the standard deviation is $\sigma = 205$. Sample size is $n = 60$.
Let $\bar{X}$ be the sample mean.
$\bar{X}$ is less than 900

$$\begin{aligned}
P(\bar{X} < 900) &= P(\bar{X} - \mu < 900 - \mu) \\
&= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{900 - \mu}{\sigma/\sqrt{n}}\right) \\
&= P\left(z < \frac{900 - 947}{205/\sqrt{60}}\right) \\
&= P\left(z < \frac{-47}{26.45}\right) \\
&= P(z < -1.78) \\
&= P(z > 1.78) \\
&= 0.5 - P(0 < z < 1.78) \\
&= 0.5 - 0.4625 \quad \text{(from Appendix B1.)} \\
&= 0.0375
\end{aligned}$$

Result

$$0.0735$$

PROBLEM STATEMENT

Chapter 8, Page 294

Step 1

(a) There are 6 possible values for each side: 1, 2, 3, 4, 5, and 6.

Fundamental counting principle (multiplication rule): If the first event could occur in m ways and the second event could occur in n ways, then the number of ways that the two events could occur in sequence is $m \cdot n$.

$$6 \cdot 6 = 36$$

Thus there are 36 different samples.

Step 2

(b) We need to list all possible samples of two observations and compute the sample mean of each of the samples.

The sample mean is the sum of all data values in the sample divided by the number of data values.

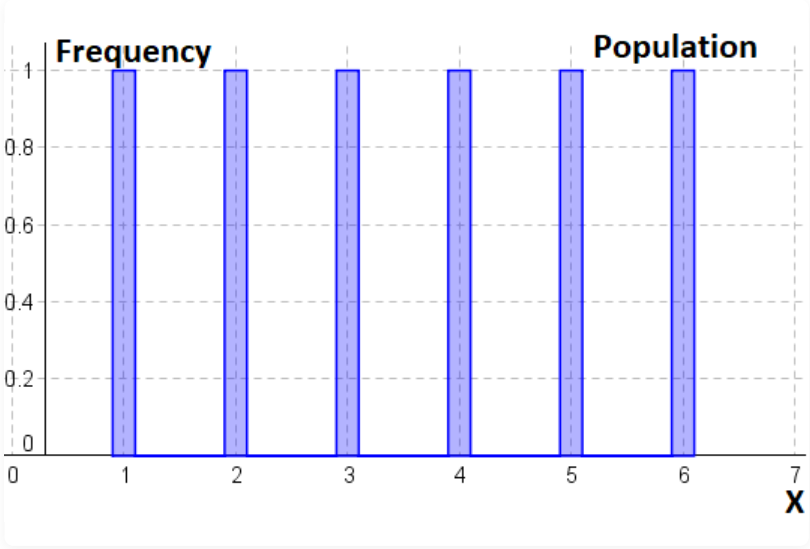
Sample (first die, second die)	Sample mean
$1, 1$	$(1+1)/2 = 1$
$1, 2$	$(1+2)/2 = 1.5$
$1, 3$	$(1+3)/2 = 2$
$1, 4$	$(1+4)/2 = 2.5$
$1, 5$	$(1+5)/2 = 3$
$1, 6$	$(1+6)/2 = 3.5$
$2, 1$	$(2+1)/2 = 1.5$
$2, 2$	$(2+2)/2 = 2$
$2, 3$	$(2+3)/2 = 2.5$
$2, 4$	$(2+4)/2 = 3$
$2, 5$	$(2+5)/2 = 3.5$
$2, 6$	$(2+6)/2 = 4$
$3, 1$	$(3+1)/2 = 2$
$3, 2$	$(3+2)/2 = 2.5$
$3, 3$	$(3+3)/2 = 3$
$3, 4$	$(3+4)/2 = 3.5$
$3, 5$	$(3+5)/2 = 4$
$3, 6$	$(3+6)/2 = 4.5$

Step 3

Sample (first die, second die)	Sample mean
$4, 1$	$(4+1)/2 = 2.5$
$4, 2$	$(4+2)/2 = 3$
$4, 3$	$(4+3)/2 = 3.5$
$4, 4$	$(4+4)/2 = 4$
$4, 5$	$(4+5)/2 = 4.5$
$4, 6$	$(4+6)/2 = 5$
$5, 1$	$(5+1)/2 = 3$
$5, 2$	$(5+2)/2 = 3.5$
$5, 3$	$(5+3)/2 = 4$
$5, 4$	$(5+4)/2 = 4.5$
$5, 5$	$(5+5)/2 = 5$
$5, 6$	$(5+6)/2 = 5.5$
$6, 1$	$(6+1)/2 = 3.5$
$6, 2$	$(6+2)/2 = 4$
$6, 3$	$(6+3)/2 = 4.5$
$6, 4$	$(6+4)/2 = 5$
$6, 5$	$(6+5)/2 = 5.5$
$6, 6$	$(6+6)/2 = 6$

Step 4

(c) We create a histogram for both the population and for all sample means in part (b), where the population values or sample mean are on the horizontal axis, the bars are equally wide, and the height of the bars is equal to the number of times the particular data value occurs.

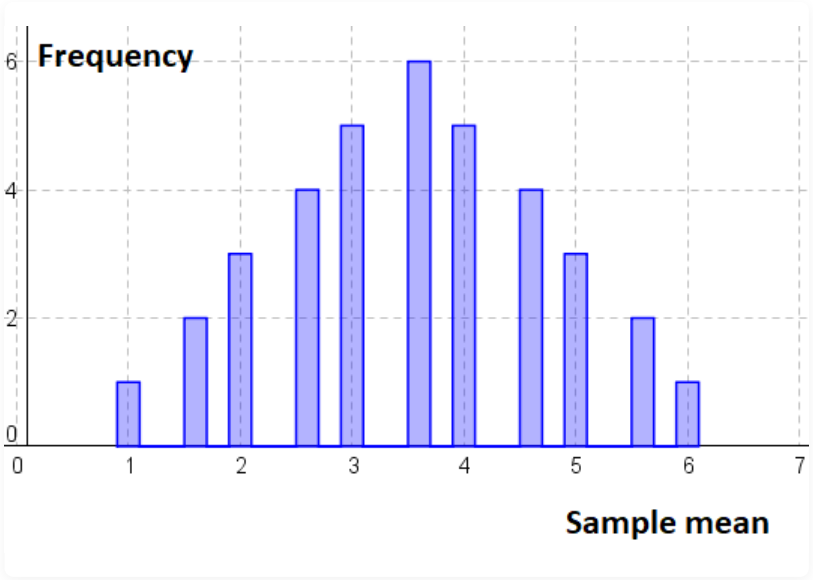


The population distribution is uniform and symmetric, because all bars have the same height.

The distribution of the sample means is symmetric and approximately bell-shaped, because the highest bar is approximately in the middle of the histogram.

Thus both distributions are symmetric.

Step 5



Step 6

(d)

Sampling distribution of the sample means

Next, we need to determine the mean and standard deviation of all sample means in part (b):

$$\begin{aligned}\mu_{\overline{X}} &= \frac{\begin{array}{r}1+1.5+2+2.5+3+3.5\\+1.5+2+2.5+3+3.5+4\\+2+2.5+3+3.5+4+4.5\\+2.5+3+3.5+4+4.5+5\\+3+3.5+4+4.5+5+5.5\\+3.5+4+4.5+5+5.5+6\end{array}}{36}\\ &= \frac{126}{36}\\ &= \frac{7}{2}\\ &= 3.5\end{aligned}$$

n is the number of values in the data set.

$$n = 36$$

The **population variance** is the sum of squared deviations from the mean divided by n:

$$\begin{aligned}\sigma^2_{\overline{X}} &= \frac{\sum(x-\overline{x})^2}{n}\\ &= \frac{(1-3.5)^2+(1.5-3.5)^2+....+(6-3.5)^2}{36}\\ &= \frac{43.75}{36}\\ &\approx 1.4583\end{aligned}$$

The **population standard deviation** is the square root of the population variance:

$$\sigma_{\overline{X}} = \sqrt{\sigma^2_{\overline{X}}} = \sqrt{1.4583} \approx 1.2247$$

Step 7

Population distribution

The population mean is the sum of all population value divided by the number of population values.

$$\mu = \frac{\sum x_i}{N} = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5$$

n is the number of values in the data set.

$$n = 6$$

The **population variance** is the sum of squared deviations from the mean divided by n:

$$\begin{aligned}\sigma^2 &= \frac{\sum(x-\overline{x})^2}{n} \\ &= \frac{(1-3.5)^2+(2-3.5)^2+....+(6-3.5)^2}{6} \\ &= \frac{17.5}{6} \\ &\approx 2.9167\end{aligned}$$

The **population standard deviation** is the square root of the population variance:

$$\sigma = \sqrt{\sigma^2} = \sqrt{2.9167} \approx 1.8708$$

We note that the population mean is equal to the mean of all sample means, while the population standard deviation is greater than the standard deviation of all sample means.

🎯 Result

(a) 36 different samples

(b) Sample means:

(c) Both distributions are symmetric.

(d) Same means, smaller standard deviation for sample means

PROBLEM STATEMENT

Chapter 8, Page 294

Step 1

(a) Selected random numbers: 45, 15, 81, 09, 39, 43, 90, 26, 06, 45, 01, 42

The given ID numbers range from 00 to 49, thus we need to eliminate all random numbers that exceed 49:

$$45, 15, 09, 39, 43, 26, 06, 45, 01, 42$$

We note that the value 45 is repeated, thus we remove one of the repeated values:

$$45, 15, 09, 39, 43, 26, 06, 01, 42$$

We only need a sample of 8, thus we only use the first 8 numbers.

$$45, 15, 09, 39, 43, 26, 06, 01$$

The corresponding states in the sample are then: Virginia, Kansas, Georgia, South Carolina, Utah, Nebraska, Connecticut, Alaska.

Step 2

(b) We need to select every sixth location with starting point 02.

Let us first determine the Id numbers that will be in the sample, which are the starting point and the starting point increased by 6 or a multiple of 6.

The given ID numbers range from 00 to 49, thus we need to eliminate all random numbers that exceed 49.

$$
\begin{aligned}
&02\\
02+06=&08\\
08+06=&14\\
14+06=&20\\
20+06=&26\\
26+06=&32\\
32+06=&38\\
38+06=&44\\
44+06=&50
\end{aligned}
$$

The selected random numbers are then 02, 08, 14, 20, 26, 32, 38, and 44. The corresponding states are: Arizona, Florida, Iowa, Massachusetts, Nebraska, North Carolina, Rhode Island, Vermont.

Result

(a) Virginia, Kansas, Georgia, South Carolina, Utah, Nebraska, Connecticut, Alaska.

(b) Arizona, Florida, Iowa, Massachusetts, Nebraska, North Carolina, Rhode Island, Vermont.

PROBLEM STATEMENT

Chapter 8, Page 294

Step 1

The mean deductible amount per employee is $\mu = \$502$ and the standard deviation is $\sigma = \$100$. Sample size of companies is $n = 60$.

Let $\bar{X}$ be the sample mean.

a. Standard error of the mean

$$\begin{aligned}\sigma/\sqrt{n} &= 100/\sqrt{60} \\ &= 100/7.75 \\ &= 12.9\end{aligned}$$

Step 2

$\textbf{b. } \bar{X}$ is between 477 and 527 }$

$$\begin{aligned}P(477 < \bar{X} < 527) &= P(477 < \bar{X} - \mu < 527 - \mu) \\ &= P\left(\frac{477-\mu}{\sigma/\sqrt{n}} < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < \frac{527-\mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(\frac{477-502}{100/\sqrt{60}} < z < \frac{527-502}{100/\sqrt{60}}\right) \\ &= P\left(\frac{-25}{12.9} < z < \frac{25}{12.9}\right) \\ &= P(-1.94 < z < 1.94)\end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-1.94 < z < 0)$, $P(0 < z < 1.94)$ and add the two probabilities.

$$\begin{aligned}P(477 < \bar{X} < 527) &= P(-1.94 < z < 0) + P(0 < z < 1.94) \\ &= P(0 < z < 1.94) + P(0 < z < 1.94) \\ &= 0.4738 + 0.4738 \ \ (\text{from Appendix B1.}) \\ &= 0.9476\end{aligned}$$

Step 3

$\textbf{c. } \bar{X}$ is between 477 and 527 }$

$$\begin{aligned} P(477 < \bar{X} < 527) &= P(477 < \bar{X} - \mu < 527 - \mu) \\ &= P\left(\frac{477 - \mu}{\sigma/\sqrt{n}} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{527 - \mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(\frac{477 - 502}{100/\sqrt{60}} < z < \frac{527 - 502}{100/\sqrt{60}}\right) \\ &= P\left(\frac{-25}{12.9} < z < \frac{25}{12.9}\right) \\ &= P(-1.94 < z < 1.94) \end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-1.94 < z < 0)$, $P(0 < z < 1.94)$ and add the two probabilities.

$$\begin{aligned} P(477 < \bar{X} < 527) &= P(-1.94 < z < 0) + P(0 < z < 1.94) \\ &= P(0 < z < 1.94) + P(0 < z < 1.94) \\ &= 0.4738 + 0.4738 \ \ (\text{from Appendix B1.}) \\ &= 0.9476 \end{aligned}$$

Step 4

$\textbf{c. } \bar{X}$ is between 492 and 512 }$

$$\begin{aligned} P(492 < \bar{X} < 512) &= P(492 < \bar{X} - \mu < 512 - \mu) \\ &= P\left(\frac{492 - \mu}{\sigma/\sqrt{n}} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{512 - \mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(\frac{492 - 502}{100/\sqrt{60}} < z < \frac{512 - 502}{100/\sqrt{60}}\right) \\ &= P\left(\frac{-10}{12.9} < z < \frac{10}{12.9}\right) \\ &= P(-0.77 < z < 0.77) \end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-0.77 < z < 0)$, $P(0 < z < 0.77)$ and add the two probabilities.

$$\begin{aligned} P(492 < \bar{X} < 512) &= P(-0.77 < z < 0) + P(0 < z < 0.77) \\ &= P(0 < z < 0.77) + P(0 < z < 0.77) \\ &= 0.2794 + 0.2794 \ \ (\text{from Appendix B1.}) \\ &= 0.5588 \end{aligned}$$

Step 5

$\textbf{d. } \bar{X}$ is greater than 550 }$

$$
\begin{aligned}
P(\bar{X} > 550) &= P(\bar{X} - \mu > 550 - \mu) \\
&= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{550 - \mu}{\sigma/\sqrt{n}}\right) \\
&= P\left(z > \frac{550 - 502}{100/\sqrt{60}}\right) \\
&= P\left(z > \frac{48}{12.9}\right) \\
&= P(z > 3.72 \\
&= 0.5 - P(0 < z < 3.72) \\
&= 0.5 - 0.5 \\
&\text{(Probability that } z \text{ lies between 0 and } z_1 \text{ is 0.5 when } z_1 \text{ greater than 3.09)} \\
&= 0
\end{aligned}
$$

Result

a, 12.9

b. 0.9476

c. 0.5588

d. 0

PROBLEM STATEMENT

Chapter 8, Page 294

Step 1

The mean number of members who have experienced a denial-of-service attack each year is $\mu = 510$ and the standard deviation is $\sigma = 14.28$ attacks. Sample of years is $n = 10$.

Let $\bar{X}$ be the sample mean.

$\textbf{a. } \bar{X}$ is greater than 600 }$

$$
\begin{aligned}
P(\bar{X} > 600) &= P(\bar{X} - \mu > 600 - \mu) \\
&= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{600 - \mu}{\sigma/\sqrt{n}}\right) \\
&= P\left(z > \frac{600 - 510}{14.28/\sqrt{10}}\right) \\
&= P\left(z > \frac{90}{4.52}\right) \\
&= P(z > 19.91) \\
&= 0.5 - P(0 < z < 9.91) \\
&= 0.5 - 0.5 \\
&\text{Probability of } z \text{ between } 0 \text{ and } z_1 \text{ is } 0.5 \text{ for } z_1 \text{ greater than } 3.09) \\
&= 0
\end{aligned}
$$

Step 2

$\textbf{b. } \bar{X}$ is between 500 and 600 }$

$$\begin{aligned} P(500 < \bar{X} < 600) &= P(500 < \bar{X} - \mu < 600 - \mu) \\ &= P\left(\frac{500 - \mu}{\sigma/\sqrt{n}} < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{600 - \mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(\frac{500 - 510}{14.28/\sqrt{10}} < z < \frac{600 - 510}{14.28/\sqrt{10}}\right) \\ &= P\left(\frac{-10}{4.52} < z < \frac{90}{4.52}\right) \\ &= P(-2.21 < z < 19.91) \end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-2.21 < z < 0)$, $P(0 < z < 19.91)$ and add the two probabilities.

$$\begin{aligned} P(500 < \bar{X} < 600) &= P(-2.21 < z < 0) + P(0 < z < 19.91) \\ &= P(0 < z < 2.21) + P(0 < z < 19.91) \\ &= 0.4864 + 0.5 \ \ (\text{from Appendix B1.}) \\ &= 0.9864 \end{aligned}$$

Step 3

$\textbf{c. } \bar{X}$ is less than 500 }$

$$\begin{aligned} P(\bar{X} < 500) &= P(\bar{X} - \mu < 500 - \mu) \\ &= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{500 - \mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(z < \frac{500 - 510}{14.28/\sqrt{10}}\right) \\ &= P\left(z < \frac{-10}{4.52}\right) \\ &= P(z < -2.21) \\ &= P(z > 2.21) \\ &= 0.5 - P(0 < z < 2.21) \\ &= 0.5 - 0.4864 \ \ (\text{from Appendix B1.}) \\ &= 0.0136 \end{aligned}$$

🎯 **Result**

a. 0

b. 0.9864

c. 0.0136

PROBLEM STATEMENT

Chapter 8, Page 295

Step 1

Mean price per gallon of regular gasoline is $\mu = \$3$ and the standard deviation is $\sigma = \$0.18$. Sample size is $n = 40$.

Let $\bar{X}$ be the sample mean.

a. Standard error of the mean

$$\begin{aligned}\sigma/\sqrt{n} &= 0.18/\sqrt{40} \\ &= 0.18/6.32 \\ &= 0.028\end{aligned}$$

$\textbf{b. $\bar{X}$ is between 2.98 and 3.02 }$

$$\begin{aligned}P(2.98 < \bar{X} < 3.02) &= P(2.98 < \bar{X} - \mu < 3.02 - \mu) \\ &= P\left(\frac{2.98-\mu}{\sigma/\sqrt{n}} < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < \frac{3.02-\mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(\frac{2.98-3}{0.18/\sqrt{40}} < z < \frac{3.02-3}{0.18/\sqrt{40}}\right) \\ &= P\left(\frac{-0.02}{0.028} < z < \frac{0.02}{0.028}\right) \\ &= P(-0.71 < z < 0.71)\end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-0.71 < z < 0)$, $P(0 < z < 0.71)$ and add the two probabilities.

$$\begin{aligned}P(2.98 < \bar{X} < 3.02) &= P(-0.71 < z < 0) + P(0 < z < 0.71) \\ &= P(0 < z < 0.71) + P(0 < z < 0.71) \\ &= 0.2611 + 0.2611 \ \ (\text{from Appendix B1.}) \\ &= 0.5222\end{aligned}$$

Step 2

$\textbf{c. } \bar{X} - \mu \text{ is less than 0.01 }$

$$
\begin{aligned}
P(\bar{X} - \mu < 0.01) &= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{0.01}{\sigma/\sqrt{n}}\right) \\
&= P\left(z < \frac{0.01}{0.18/\sqrt{40}}\right) \\
&= P\left(z < \frac{0.01}{0.028}\right) \\
&= P(z < 0.36) \\
&= P(z > 0.36) \\
&= 0.5 - P(0 < z < 0.36) \\
&= 0.5 - 0.1406 \ \text{(from Appendix B1.)} \\
&= 0.3594
\end{aligned}
$$

Step 3

$\textbf{d. } \bar{X} \text{ is greater than 3.08 }$

$$
\begin{aligned}
P(\bar{X} > 3.08) &= P(\bar{X} - \mu > 3.08 - \mu) \\
&= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{3.08 - \mu}{\sigma/\sqrt{n}}\right) \\
&= P\left(z > \frac{3.08 - 3}{0.18/\sqrt{40}}\right) \\
&= P\left(z > \frac{0.08}{0.028}\right) \\
&= P(z > 2.86) \\
&= 0.5 - P(0 < z < 2.86) \\
&= 0.5 - 0.4979 \ \text{(from Appendix B1.)} \\
&= 0.021
\end{aligned}
$$

🎯 Result

a. 0.028

b. 0.5222

c. 0.3594

d. 0.021

PROBLEM STATEMENT

Chapter 8, Page 295

Step 1

Mean pairs of sports shoes bought per year is $\mu = 6.5$ and the standard deviation is $\sigma = 2.1$. Sample size is $n = 81$.

Let $\bar{X}$ be the sample mean.

a. Standard error of the mean

$$\begin{aligned}\sigma/\sqrt{n} &= 2.1/\sqrt{81} \\ &= 2.1/9 \\ &= 0.23\end{aligned}$$

$\textbf{b. $\bar{X}$ is between 6 and 7 }$

$$\begin{aligned}P(6 < \bar{X} < 7) &= P(6 < \bar{X} - \mu < 7 - \mu) \\ &= P\left(\frac{6-\mu}{\sigma/\sqrt{n}} < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < \frac{7-\mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(\frac{6-6.5}{2.1/\sqrt{81}} < z < \frac{7-6.5}{2.1/\sqrt{81}}\right) \\ &= P\left(\frac{-0.5}{0.23} < z < \frac{0.5}{0.23}\right) \\ &= P(-2.17 < z < 2.17)\end{aligned}$$

The two points are on the different sides of the mean, so calculate $P(-2.17 < z < 0)$, $P(0 < z < 2.17)$ and add the two probabilities.

$$\begin{aligned}P(6 < \bar{X} < 7) &= P(-2.17 < z < 0) + P(0 < z < 2.17) \\ &= P(0 < z < 2.17) + P(0 < z < 2.17) \\ &= 0.4850 + 0.4850 \ \ (\text{from Appendix B1.}) \\ &= 0.97\end{aligned}$$

Step 2

$\textbf{c. } \bar{X} - \mu \text{ is less than 0.25 }$

$$
\begin{aligned}
P(\bar{X} - \mu < 0.25) &= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{0.25}{\sigma/\sqrt{n}}\right) \\
&= P\left(z < \frac{0.25}{2.1/\sqrt{81}}\right) \\
&= P\left(z < \frac{0.25}{0.23}\right) \\
&= P(z < 1.09) \\
&= 0.5 + P(0 < z < 1.09) \\
&= 0.5 + 0.3621 \ \ (\text{from Appendix B1.}) \\
&= 0.8621
\end{aligned}
$$

Step 3

$\textbf{d. } \bar{X} \text{ is greater than 7 }$

$$
\begin{aligned}
P(\bar{X} > 7) &= P(\bar{X} - \mu > 7 - \mu) \\
&= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > \frac{7 - \mu}{\sigma/\sqrt{n}}\right) \\
&= P\left(z > \frac{7 - 6.5}{2.1/\sqrt{81}}\right) \\
&= P\left(z > \frac{0.5}{0.23}\right) \\
&= P(z > 2.17) \\
&= 0.5 - P(0 < z < 2.17) \\
&= 0.5 - 0.4850 \ \ (\text{from Appendix B1.}) \\
&= 0.015
\end{aligned}
$$

🎯 Result

a. 0.23

b. 0.97

c. 0.8621

d. 0.015

PROBLEM STATEMENT

Chapter 8, Page 295

Step 1

Let us select the 11th row of the random numbers table: $\text{\color{#4257b2}Note: Choosing a different row will result in a different sample\color{default}}\

82173 26921 28472 98958 07960 66124 89731 95069 18625 92405

97594 25168 89178 68190 05043 17407 48201 83917 11413 72920

73881 67176 93504 42636\

We select the first 3-digit number. If the number is between 001 and 105, then we select the corresponding location. If the number is greater than 105, then we ignore the number and select the next 3-digit number.\

Repeat until 10 homes were selected.\

The 10 randomly selected numbers are then: 079, 066, 069, 059, 005, 043, 074, 018, 073, 044.\

When the number is x, then we select the xth selling price in the data set. The corresponding data values of the selling price are then: 221.1, 236.8, 192.2, 207.5, 240, 172.4, 213.6, 188.3, 242.1, 251.4.}$

Step 2

Mean and standard deviation sample

The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned}\overline{x} &= \frac{\sum_{i=1}^{n} x_i}{n} \\ &= \frac{\begin{array}{c}221.1 + 236.8 + 192.2 + 207.5 + 240 \\ +172.4 + 213.6 + 118.3 + 242.1 + 251.4\end{array}}{10} \\ &= \frac{2095.4}{10} \\ &= 209.54\end{aligned}$$

n is the number of values in the data set.

$$n = 10$$

The **sample variance** is the sum of squared deviations from the mean divided by $n - 1$:

$$\begin{aligned}s^2 &= \frac{\sum(x - \overline{x})^2}{n - 1} \\ &= \frac{(221.1 - 209.54)^2 + ... + (251.4 - 209.54)^2}{10 - 1} \\ &= \frac{6278.804}{9} \\ &= 697.6449\end{aligned}$$

The **sample standard deviation** is the square root of the sample variance:

$$s = \sqrt{s^2} = \sqrt{697.6449} \approx 25.7085$$

Step 3

Mean and standard deviation population

The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned}\overline{x} &= \frac{\sum_{i=1}^{n} x_i}{n} \\ &= \frac{220.9 + 294.5 + ... + 177.1}{105} \\ &= \frac{23215.8}{105} \\ &\approx 221.1029\end{aligned}$$

n is the number of values in the data set.

$$n = 105$$

The **population variance** is the sum of squared deviations from the mean divided by n:

$$\begin{aligned}\sigma^2 &= \frac{\sum(x-\mu)^2}{n} \\ &= \frac{(220.9 - 221.1029)^2 + ... + (177.1 - 221.1029)^2}{104} \\ &= \frac{230767.5891}{104} \\ &= 2218.9191\end{aligned}$$

The **population standard deviation** is the square root of the population variance:

$$\sigma = \sqrt{\sigma^2} = \sqrt{2218.9191} \approx 46.8806$$

Step 4

Likelihood

Since the population distribution is normal, the sampling distribution of the sample mean $\overline{x}$ is also normal.

The sampling distribution of the sample mean $\overline{x}$ has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

The z-score is the value decreased by the mean, divided by the standard deviation:

$$z = \frac{\overline{x} - \mu_{\overline{x}}}{\sigma_{\overline{x}}} = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{209.54 - 221.1029}{46.8806/\sqrt{10}} \approx -0.78$$

Determine the corresponding probability using the normal probability table in the appendix.

$P(0 < Z < 0.78)$ is given in the row starting with 0.7 and in the column starting with .08 of the standard normal probability table in the appendix.

$$\begin{aligned} P(\overline{X} \geq 209.54) &= P(Z > -0.78) \\ &= P(Z < 0.78) \\ &= P(Z < 0) + P(0 < Z < 0.78) \\ &= 0.5000 + 0.2823 \\ &= 0.7823 \\ &= 78.23\% \end{aligned}$$

Result

Answers could vary

PROBLEM STATEMENT

Chapter 8, Page 295

Step 1

Given:

$$\mu = \text{Mean} = 2.25$$
$$\sigma = \text{Standard deviation} = 0.70$$
$$n = \text{Sample size} = 30$$

Mean sample

The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned}\overline{x} &= \frac{\sum_{i=1}^{n} x_i}{n} \\ &= \frac{1.46 + 1.92 + 1.58 + \ldots + 3.72}{30} \\ &= \frac{73.43}{30} \\ &\approx 2.4477\end{aligned}$$

Step 2

Likelihood

Since the population distribution is normal, the sampling distribution of the sample mean $\overline{x}$ is also normal.

The sampling distribution of the sample mean $\overline{x}$ has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$.

The z-score is the value decreased by the mean, divided by the standard deviation:

$$z = \frac{\overline{x} - \mu_{\overline{x}}}{\sigma_{\overline{x}}} = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{2.4477 - 2.25}{0.70/\sqrt{30}} \approx 1.55$$

Determine the corresponding probability using the normal probability table in the appendix.

$P(0 < Z < 1.55)$ is given in the row starting with 1.5 and in the column starting with .05 of the standard normal probability table in the appendix.

$$\begin{aligned} P(\overline{X} \geq 2.4477) &= P(Z > 1.55) \\ &= 1 - P(Z < 1.55) \\ &= 1 - P(Z < 0) - P(0 < Z < 1.55) \\ &= 1 - 0.5000 - 0.4394 \\ &= 0.0606 \\ &= 6.06\% \end{aligned}$$

Result

$$0.0606 = 6.06\%$$

PROBLEM STATEMENT

Chapter 8, Page 295

Step 1

Given:

$$\mu = \text{Mean} = 455$$
$$n = \text{Sample size} = 80$$

Mean and standard deviation sample

The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned}\overline{x} &= \frac{\sum_{i=1}^{n} x_i}{n} \\ &= \frac{329 + 503 + 505 + \ldots + 449}{80} \\ &= \frac{36023}{80} \\ &\approx 450.2875\end{aligned}$$

n is the number of values in the data set.

$$n = 80$$

The **sample variance** is the sum of squared deviations from the mean divided by $n - 1$:

$$\begin{aligned}s^2 &= \frac{\sum(x - \overline{x})^2}{n - 1} \\ &= \frac{(329 - 450.2875)^2 + \ldots + (449 - 450.2875)^2}{80 - 1} \\ &= \frac{227692.3875}{79} \\ &= 2882.1821\end{aligned}$$

The **sample standard deviation** is the square root of the sample variance:

$$s = \sqrt{s^2} = \sqrt{2882.1821} \approx 53.6860$$

Step 2

Likelihood

Since the population distribution is normal, the sampling distribution of the sample mean $\overline{x}$ is also normal.

The sampling distribution of the sample mean $\overline{x}$ has mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$ (we will estimate the population standard deviation σ as $s = 53.6860$).

The z-score is the value decreased by the mean, divided by the standard deviation:

$$z = \frac{\overline{x} - \mu_{\overline{x}}}{\sigma_{\overline{x}}} = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{450.2875 - 455}{53.6860/\sqrt{80}} \approx -0.79$$

Determine the corresponding probability using the normal probability table in the appendix.

$P(0 < Z < 0.79)$ is given in the row starting with 0.7 and in the column starting with .09 of the standard normal probability table in the appendix.

$$\begin{aligned} P(\overline{X} \leq 450.2875) &= P(Z < -0.79) \\ &= P(Z > 0.79) \\ &= 1 - P(Z < 0.79) \\ &= 1 - P(Z < 0) - P(0 < Z < 0.79) \\ &= 1 - 0.5000 - 0.2852 \\ &= 0.2148 \\ &= 21.48\% \end{aligned}$$

Result

$$0.2148 = 21.48\%$$